



Combustion and pyrolysis of *iso*-butanol: Experimental and chemical kinetic modeling study

Shamel S. Merchant^a, Everton Fernando Zanoelo^b, Raymond L. Speth^a, Michael R. Harper^{a,1}, Kevin M. Van Geem^c, William H. Green^{a,*}

^a Department of Chemical Engineering, Massachusetts Institute of Technology, Cambridge, MA 02139, United States

^b Federal University of Paraná (UFPR), Curitiba, PR, Brazil

^c Laboratory for Chemical Technology, Univ. Ghent, Ghent, Belgium

ARTICLE INFO

Article history:

Received 12 October 2012

Received in revised form 25 March 2013

Accepted 26 April 2013

Available online xxxx

Keywords:

iso-Butanol

Pyrolysis

Combustion

Reaction mechanism

ABSTRACT

The first reaction mechanism for *iso*-butanol (372 species and 8723 reversible elementary reactions) pyrolysis and combustion that includes pressure dependent kinetics and proposes reaction pathways to soot precursors has been automatically generated using the open-source software package RMG. High-pressure reaction rate coefficients for important hydrogen abstraction reactions from *iso*-butanol by hydrogen, methyl and HO₂ were calculated using quantum chemistry at the CBS-QB3 level. The mechanism was validated with recently published *iso*-butanol combustion experiments as well as new pyrolysis speciation data under diluted and undiluted conditions from 900 to 1100 K at 1.72 atm representative of fuel rich combustion conditions. Sensitivity and rate of production analysis revealed that the overall good agreement for the pyrolysis species, and in particular for the soot precursors like benzene, toluene and 1,3-cyclopentadiene, depends strongly on pressure dependent reactions involving the resonantly stabilized *iso*-butenyl radical. Laminar flame speed, opposed flow diffusion flame speciation profiles, and autoignition are also well-captured by the model. The agreement with speciation profiles for the jet-stirred reactor could be improved, in particular for temperatures lower than 850 K. Flux and sensitivity analysis for *iso*-butanol consumption revealed that this is primarily caused by uncertainty in *iso*-butanol + OH, *iso*-butanol + HO₂ and the low temperature peroxy chemistry rates. Further theoretical and quantum chemical studies are needed in understanding these rates to completely predict the combustion behavior of *iso*-butanol using detailed chemistry.

© 2013 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

1. Introduction

The demand for a clean, renewable biofuel increases as new benchmarks are legislated amid increased pressure to reduce the world's dependence on fossil fuels for energy and chemicals. Bio-butanol is in pole position to become a major game changer in the 21st century [1]. It is considered an advanced biofuel – superior to ethanol in almost every way – and convertible to jet fuel and biogasoline [2]. Not only are the fuel properties of butanol better than ethanol, butanol is also considered a ‘greener’ fuel than ethanol [3]. One of the remaining unanswered questions is which of the four butanol isomers (*n*-butanol, *sec*-butanol, *iso*-butanol, and *tert*-butanol) should be favored in the coming decades.

Although today the butanol isomers are largely produced from petrochemicals, all isomers, apart from tertiary butanol, can be

developed independently in renewable forms [4] to be marketed as biobutanol at competitive prices to oil based fuels. *n*-Butanol was initially considered to be the logical favorite to replace ethanol because of its mature production technology via the well-known acetone–butanol–ethanol process by anaerobic fermentation of sugar substrates using solventogenic clostridia [5]. However, this technology is handicapped in competitiveness by high energy cost due to low concentrations of butanol produced and significant volumes of water processed, even with new clostridia strains [2]. That is why in recent years *iso*-butanol has gradually become favored by industry. One of the main advantages of the *iso*-butanol technology is that it can be produced at adapted ethanol plants from corn and other grains or sugarcane and even from cellulose, as demonstrated by among others Gevo Inc. [6,7]. With diminishing subsidies on conventional fermentation ethanol, producers looking for higher margin derivatives are potential candidates for switching to *iso*-butanol. Looking at the fuel properties of both isomers the differences are relatively small with the exception of *iso*-butanol having a higher octane rating (motor octane number of 89)

* Corresponding author.

E-mail address: whgreen@mit.edu (W.H. Green).

¹ Present address: Exxon Mobil Research and Engineering Company, Rte. 22 East, Annandale, NJ 08801, United States.

compared to *n*-butanol (motor octane number of 78), this gives it a slightly higher value as gasoline blend stock.

The strong interest from industry in butanol as a potential replacement for ethanol in combustion engines has resulted in many recent combustion studies of the different butanol isomers. Only a brief summary, focusing on *iso*-butanol, is presented in this work. *iso*-Butanol has received considerable attention from the engine community with studies performed in spark-ignition engines [8,9] and in compression-ignition engines [10,11]. These studies are essential for assessing the feasibility and performance of *iso*-butanol in practical systems and are complementary with more fundamental combustion science research. The latter include experimental studies on premixed laminar flame propagation [12,13], premixed laminar flame structure [14], non-premixed laminar diffusion flame structure [13,15,16], shock tube and rapid compression machine ignition delay [17–19], and jet-stirred reactor oxidation chemistry [20].

All this recent experimental work has certainly helped the development and validation of detailed chemical kinetic models for *iso*-butanol, such as those presented by [13,16,21–23]. These fundamental kinetic models have the ambition to be predictive in nature with the ultimate goal of advancing design and optimization of practical engine systems through coupling with computational fluid dynamic (CFD) models. However, despite the recent leaps in butanol kinetic modeling, a comprehensive model for *iso*-butanol pyrolysis and combustion – valid at the temperature and pressure ranges found in practical engines – is still lacking. Moreover the experimental dataset for *iso*-butanol is far from complete. For example, although pyrolysis reactions are an essential part of any combustion mechanism [24], prior to this work accurate pyrolysis data for *iso*-butanol have not been available. These pyrolysis reactions play an important role in areas of fuel rich conditions in internal combustion engines, e.g. in case of incomplete mixing.

The goal of the present study is to develop a detailed kinetic mechanism for *iso*-butanol pyrolysis and combustion building on the mechanisms developed for other isomers in earlier work [32,54]. This study differentiates itself from other *iso*-butanol chemical kinetic model studies by consistently including pressure dependent kinetics, with most rate coefficients from high-pressure limit rate rules built into the open source mechanism generator RMG [25]. Most rate rules originate from literature but for the most important reactions in our system new rate rules have been determined from quantum calculations. The mechanism is tested against a large number of new and recently published data including – pyrolysis product profiles in plug flow reactor, jet-stirred reactor mole fraction profiles, opposed flow diffusion flame mole fraction profiles, laminar flame speeds at different pressures and autoignition delays in rapid-compression machine and high-pressure shock tube experiments. Furthermore, this study attempts to guide future *iso*-butanol pyrolysis and combustion chemistry research by elucidating those reactions that are the most significant and uncertain for the experimental conditions that have been studied.

2. Experiment and computational procedures

2.1. Pyrolysis experiments

The experimental set-up has been described in detail elsewhere [26–28] and therefore only a brief description based on Djokic et al. [28] and Pyl et al. [27] will be given here. A schematic overview of the experimental set-up as well as a more extensive description are given in the [Supplementary information](#) section. It consists of three parts: the feed section, the furnace/reactor section and the

analysis section. *iso*-Butanol (Acros Organics, Belgium, purity 99.5%+, used as obtained) is pumped towards an evaporator kept at 523 K using a peristaltic pump (Heidolph, Germany). The mass flow rate of the feed is calibrated using an electronic balance. The diluent, N₂ (Air Liquide, Belgium, purity 99.999%), is heated separately to the same temperature. Both the evaporators/heaters and the subsequent mixer are electrically heated and filled with quartz beads, enabling a smooth evaporation of the feed and uniform mixing of the feed and diluent. The flow rate of the latter is controlled by a coriolis mass flow controller (Bronkhorst, The Netherlands). Typical experimental conditions for diluted and undiluted *iso*-butanol are provided in [Table 1](#). The overall residence time varies between 1.1 s and 0.57 s. Working with fixed mass flow rates for diluted and undiluted conditions respectively results in a difference in residence time at higher temperatures of 0.18 s because of a difference in molar flow rate due to the decomposition of *iso*-butanol into predominantly species with lower molecular weight. Due to this increase in residence time, the conversion for the diluted experiments is higher for an identical temperature profile in the plug flow reactor.

The reactor is a 1.475 m long, 6 mm internal diameter tube, made of Incoloy 800HT (Ni, 30–35; Cr, 19–23; and Fe, >39.5 wt%). Prior to its use in these experiments, the reactor tube was passivated to suppress potential catalytic activity of the wall. Control experiments with a second reactor having a two times larger surface to volume ratio showed that the influence of the wall on conversion and product selectivities was negligible. The reactor is placed vertically in an electrically heated rectangular furnace. In all experiments discussed in this work, the reactor is operated nearly isothermally, i.e. with a steep temperature increase at the inlet and a steep temperature drop at the reactor outlet. Eight thermocouples along the reactor monitor the process gas temperature at different positions. The temperature profiles of experiments are provided in [Supplementary information](#). The pressure in the reactor is controlled by a downstream back pressure regulator. Two manometers, positioned at the inlet and outlet of the reactor, record the coil inlet pressure (CIP) and the coil outlet pressure (COP), respectively. The pressure drop over the reactor was found to be negligible. During a typical 6-h operation the pressure drop did not increase noticeably since the amount of coke deposited on the reactor inner wall was found to be marginal. Even under the most severe conditions ($T > 1073$ K) no significant pressure drop was observed, with the pressure remaining constant in the coil at 1.7 bar. No entrained coke particles were collected.

The analysis section of the pyrolysis set-up is similar to the one discussed previously [29,30]. A detailed description of the columns used and analysis procedure can be found in the [Supporting information](#). More than 100 different components are observed and identified using the on-line comprehensive 2D gas chromatograph (GC × GC) equipped with a flame ionization detector (FID) and a Time-of-Flight Mass Spectrometer (TOF-MS). [Figure 1](#) shows part of the GC × GC chromatogram during one of the *iso*-butanol experiments. The settings and the characteristics of the GC × GC can be found elsewhere [27]. The increased separation power of GC × GC makes it very suitable for analyzing complex effluents containing

Table 1

Overview of experimental conditions for the pyrolysis of *iso*-butanol in the bench scale set-up.

	Diluted	Undiluted
T_{inl} (K)	550–660	550–660
T_{avg} (K)	913–1093	913–1093
Outlet pressure (10 ⁵ Pa)	1.72	1.72
Hydrocarbon flow rate (g s ^{−1})	1.666×10^{-2}	6.666×10^{-2}
Nitrogen flow rate (g s ^{−1})	1.889×10^{-2}	0
Conversion (%)	15–98	15–94

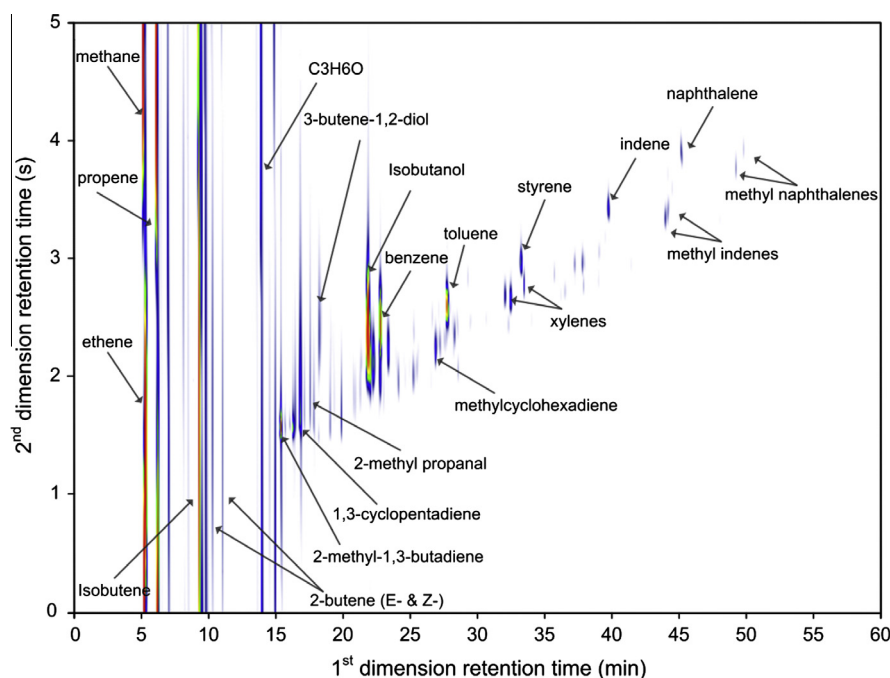


Fig. 1. GC $\times$ GC-FID chromatogram of the on-line sampled undiluted *iso*-butanol pyrolysis effluent at $T_{\text{avg}} = 1053 \text{ K}$ and $F_{\text{iso-butanol}} = 6.66 \times 10^{-2} \text{ g s}^{-1}$.

oxygenates. The 1st dimension separation is based on volatility, the 2nd on polarity. Major products formed during the cracking of *iso*-butanol include hydrogen, CO, methane, formaldehyde, ethene, propene, *iso*-butene, propenal, butanol and ethanol. Minor products, such as methanol, *iso*-butanal, ketene, acetaldehyde, acetone, 1 and 2-butene, and 1,3-butadiene are also found in the effluent. At higher conversions significant amounts of 1,3-cyclopentadiene, benzene, toluene, xylenes, and ethylbenzene, and even traces of naphthalene, methyl naphthalene and biphenyl are detected. Peak identification and integration of the FID chromatogram is performed by a commercial integration package, GC-Image (Zoex Corp.), using the information obtained from GC $\times$ GC TOF-MS analysis. For each studied condition, at least three repeat analyses were performed. For all runs the mass balances were within the range of 95–105%. Deviations are attributed to uncertainties in the mass flow rates of *iso*-butanol and the internal standard (N_2). Subsequently, the component weight fractions were normalized to 100 wt%, in order to enable straightforward interpretation of the observed trends. After normalization of the total mass balance, elemental balances (C, H, O) were verified. The estimated uncertainty of the measured product yields is 5% rel in line with previous work [28,29]. Uncertainty is lower for components with higher mole fractions.

2.2. Reactor models

2.2.1. Pyrolysis

For simulating the pyrolysis experiments in the pyrolysis set-up, the reaction network was implemented in Chemkin-Pro [31]. The laminar flow reactor was simulated first using Chemkin's cylindrical shear flow option to investigate the importance of radial temperature and concentration gradients for this set-up. In line with the model results obtained for the three other butanol isomers [26,32] and mixtures of butanol [30] minor radial concentration and temperature non-uniformities are computed, indicating that the assumption of using a plug flow reactor model for this reactor is indeed valid [33]. Based on the characteristic times and nondimensional numbers [34–36] pertinent to our tubular

flow reactor the validity of the plug flow treatment of tubular reactor data has been further verified and these results show that the plug flow treatment is valid, see the [Supplementary information](#). Hence, we have employed the Chemkin-Pro plug flow reactor (PFR) module for modeling the tubular reactor. The PFR module was supplied with the measured temperature and pressure profile, and the reactor dimensions used were those discussed in Section 2.1. The default absolute and relative tolerances were sufficient for convergence.

2.2.2. Doped methane diffusion flame

The doped methane flame data of McEnally and Pfefferle [15] were simulated using the reactor model constructed and utilized by Sharma et al. [37], however significant changes have been made in the computational method used for solving the problem. The Sharma et al. [37] solver could not handle the stiff system presented by the butanol model which has been overcome in the current solver. Only a brief description of the reactor model is given in this work; please see [37] for a more detailed explanation.

The doped methane flame was modeled as a steady-state system, using an axisymmetric geometry. In general, solving this problem requires solving for the temperature, axial and radial velocities, and species concentration profiles as a function of r and z . Bennett et al. [38] solved this problem for an undoped methane diffusion flame using the same reactor geometry as McEnally and Pfefferle [15]. Additionally McEnally and Pfefferle have demonstrated that for the small amount of dopant used, the centerline temperature profile exhibits no significant change between the undoped- and doped-flame, within the experimental uncertainty of 65 K for the gas temperature [39]. Hence, we used the temperature profile obtained in the numerical solution of the undoped methane flame as the temperature profile for the doped-methane flame. Furthermore, since the temperature profile was not greatly affected, we assumed the velocity profiles would not be affected significantly either. With the temperature and velocity profiles known, the reactor model only needs to solve for the species concentration profiles in the doped methane flame.

The form of the species conservation equation is given in the following equation:

$$\rho v_z \frac{\partial Y_i}{\partial z} + \rho v_r \frac{\partial Y_i}{\partial r} = \gamma_i MW_i - \frac{1}{r} \frac{\partial}{\partial r} (r \rho v_{r,i} Y_i) - \frac{\partial}{\partial z} (\rho v_{z,i} Y_i) \quad (1.1)$$

ρ is the total density, v_z is the axial velocity, Y_i is the mass fraction of species i , v_r is the radial velocity, γ_i is the molar rate of production of species i , MW_i is the molecular weight of species i , and the radial and axial diffusive flux terms are given in Eqs. (1.2) and (1.3), respectively. D_i is the mixture-averaged diffusion coefficient for species i .

$$v_{r,i} Y_i = -D_i(Y) \frac{\partial Y_i}{\partial r} \quad (1.2)$$

$$v_{z,i} Y_i = -D_i(Y) \frac{\partial Y_i}{\partial z} \quad (1.3)$$

As the Peclet number on the centerline exceeds 70 we neglect the axial diffusion term and solve the species conservation equation using methods of lines. Moving the radial convective term from the left-hand side of Eq. (1.1) to the right-hand side isolates all r -dependent terms to the right-hand side of the equation, resulting in the following equation:

$$\rho v_z \frac{\partial Y_i}{\partial z} = \gamma_i MW_i - \rho v_r \frac{\partial Y_i}{\partial r} - \frac{1}{r} \frac{\partial}{\partial r} (r \rho v_{r,i} Y_i) \quad (1.4)$$

Using second-order centered finite differences to approximate the diffusion term and first-order upwind differences to approximate the radial convection term of Eq. (1.4) reduces the overall problem from a partial differential equation to a first-order ordinary differential equation initial-value problem (ODE-IVP). This problem was solved using a modified form of Strang splitting [40] which reduces errors for some stiff problems [41]. The reaction terms were integrated using the α -QSS (quasi steady state) method [42]; the convection terms were integrated using the variable-order Adams–Bashforth method as implemented by CVODE [43]; and the diffusion terms were integrated using a second-order backward differentiation formula (BDF) method. Thermodynamic properties, transport coefficients and kinetic rates were evaluated using Cantera [44].

The $z = 0$ initial boundary conditions for the doped methane flame is all species mole fractions are zero except for the species CH_4 , N_2 , O_2 , Ar, and *iso*-butanol. There are two boundary conditions in the r -direction for each species. Since the geometry of the problem is assumed to be axisymmetric, $dC_i/dr = 0$ at $r = 0$ for all species. The other boundary condition we impose in the r -direction is zero species flux ($dC_i/dr = 0$) at the edge of the reactor model, 54-mm from the center of the burner; the burner has an outer diameter of 6.35-mm.

2.2.3. Jet-stirred reactor

The mechanism was validated against the jet-stirred reactor (JSR) data measured by Togbé et al. [20]. The reactor was simulated using the Perfectly Stirred Reactor (PSR) module in Chemkin-Pro [31]. The reactor is solved as an isothermal system with volume of 30.5 cm^3 . The JSR is solved for feed mole fraction of *iso*-butanol 0.1% (0.15% for $\Phi = 4$), the equivalence ratios simulated are 0.5, 1, 2 and 4 with N_2 as bath gas. The pressure for all simulations is 10 atm spanning the temperature range of 700–1250 K with a residence time of 0.7 s. The absolute and relative tolerance of the solver was set to 1×10^{-16} and 1×10^{-8} respectively; the absolute and relative tolerance for pseudo time stepping was also set to 1×10^{-16} and 1×10^{-8} respectively.

2.2.4. Laminar burning velocities

This study also reports the premixed laminar flame velocities as measured (and extrapolated to zero strain) by Veloo and

Egolfopoulos [12] and Liu et al. [13]. Here these velocities were simulated using the PREMIX laminar flame code module in Chemkin-Pro [31]. The final temperature profile was set at the adiabatic flame temperature and unburnt gas temperature was set to 343 K and 353 K for measurements by Veloo and Liu et al. respectively. The measurements by Veloo and Egolfopoulos were only at atmospheric pressure, while Liu et al. measured the flame speed at 1 atm and 2 atm.

The PREMIX flame code was solved with initial grid points of 200 adaptable to maximum of 1500. The simulations accounted for thermal diffusion (i.e. Soret effect) and used mixture averaged diffusivities. The adaptive grid control based on solution curvature and gradient were reduced to 0.2 in order to obtain converged results. The absolute and relative tolerances of the solver are set to 1×10^{-6} and 1×10^{-3} .

2.2.5. Opposed flow diffusion flame

The mechanism was also validated against the opposed-flow diffusion data of Grana et al. [16] using the Premixed Opposed-flow flame module in Chemkin-MFC [45]. The energy equation is solved in the simulation with the experimental temperature profile provided as an initial guess to the solver. The default settings of the Chemkin solver are used for the simulation i.e. cylindrical coordinates, windward differencing scheme, thermal diffusion and mixture averaged transport properties. The absolute and relative tolerances of the solver are set to 1×10^{-9} and 1×10^{-4} . The adaptive grid control based on solution gradient and curvature were both set to 0.5. Lowering the gradient and curvature setting did not have any noticeable effect on the solution.

The simulated conditions for fuel inlet stream are inlet velocity of 24.8 cm/s and inlet temperature of 353 K with an initial mole fraction of 0.14 for *iso*-butanol and 0.86 for N_2 . The oxidizer stream has an inlet velocity of 25 cm/s and temperature of 298 K with mole fractions of 0.21 and 0.79 for O_2 and N_2 respectively.

2.2.6. Shock tube experiments

The shock tube experiments [17] were modeled in Chemkin-Pro [31], using the Closed Homogenous Batch reactor module. The reactor was solved as an adiabatic system with constant volume. The default absolute and relative tolerances were used for the simulations, including the one where the time-dependent sensitivities were calculated. The ignition delay was calculated as the time required to reach half of maximum OH concentration, and was found to be reasonably close to other metrics used in literature (i.e. the time of steepest rise in pressure and the time at which the product $[\text{C}_2\text{H}] \times [\text{O}]$ reaches its maximum).

3. Reaction mechanism generation

3.1. Automatic reaction mechanism generation and refinement

The chemical kinetic model presented in this paper is constructed using the automated “Reaction Mechanism Generator” (RMG) software package [25], originally developed by Song [46]. RMG uses a rate based [47] algorithm, that constructs a valid reaction network at given conditions and user defined tolerances. RMG estimates the high pressure kinetics using over 40 RMG-defined reaction templates (e.g. H-abstraction, radical recombination, β -scissions), the fall-off effects are estimated using Modified Strong Collision (MSC) approach of Chang et al. [48] or Reservoir State method (RS) developed by Green and Bhatti [49]. The pressure dependent kinetics for this mechanism have been estimated by Modified Strong Collision. It was found after extensive analysis that MSC provides more robust estimates than RS when considered over the entire temperature and pressure range. The RS estimates

were found to be much poorer than MSC when compared to Chemically Significant Eigenvalue (CSE) analysis [50] when a well in the pressure dependent network was shallow. In these cases the rates from RS at low pressure or high temperatures were unphysically orders of magnitudes different from the CSE value and would lead to extreme stiffness in the kinetic equations. (As discussed in literature, RS is quite accurate for deep wells and at low temperatures [49,51].)

The thermochemistry of most species in the mechanism are estimated using Benson's group additivity scheme [52]. The transport properties like Lennard-Jones parameters for species are estimated using empirical correlations based on species critical properties i.e. temperature and pressure, which are estimated using a group-additivity scheme [53]. Whenever the user has better knowledge of reaction rate expressions and species thermochemistry or transport properties, these can be supplied externally to RMG in the form of a library and will be used instead of RMG estimates. This is advantageous as the uncertainty associated with the rate coefficient estimated by RMG (from functional group analogy) can be greater than an order of magnitude especially in cases where no closely analogous literature rate is available. A RMG simulation can be seeded with an entire reaction mechanism which will then be expanded using the rate based algorithm until it satisfies the termination criterion for the simulation; in this manner a previous model can be expanded for different starting species and initial conditions.

The current *iso*-butanol mechanism is made by using the previous butanol model that has been developed for *n*-butanol, sec-butanol and tert-butanol [32,54], as its seed mechanism. Briefly the seed consists of thermochemistry parameters and rate coefficients from previous butanol model (281 species and 3600 reactions) for *n*-, sec- and tert-butanol isomers. The small molecule chemistry in the seed originated from GRI-Mech 3.0 [55] and the Marinov ethanol mechanism [56] with updates as mentioned in Refs. [26,54]. The conditions for the RMG simulation used to grow the mechanism are:

- *iso*-Butanol pyrolysis: The simulation was run at a pressure of 1.5 atm over a temperature range of 900–1200 K (in steps of 100 K) for initial mass fraction of 1 for *iso*-butanol and diluted case with mass fraction of 0.33 for *iso*-butanol and 0.67 N₂. The termination criterion was 0.99% conversion of fuel in all cases.
- *iso*-Butanol oxidation: These simulations were run at pressures of 1 and 10 bars over a temperature range of 700–2100 K (in steps of 200 K) for equivalence ratio Φ of 0.5, 1, 2 and 4. These conditions cover most of the operating regime of the experiments discussed below in this paper.

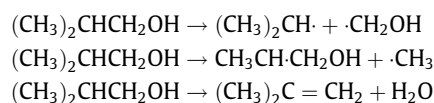
The following additional updates have been made to the model reported in [54].

3.1.1. Updated H₂–O₂ mechanism

The current reaction model uses the updated H₂–O₂ mechanism as proposed by Burke et al. [57]. The reactions with H₂ and O₂ often show up as the most sensitive reactions in most simulations, hence it is important to use the best available parameters for these reactions. The model by Burke et al. updated a number of rate constants; some of important ones are $\text{H} + \text{O}_2(+\text{M}) = \text{HO}_2(+\text{M})$, $\text{H} + \text{O}_2 = \text{OH} + \text{O}$, $\text{HO}_2 + \text{H} = \text{H}_2 + \text{O}_2$, $\text{H}_2\text{O} + \text{M} = \text{H} + \text{OH} + \text{M}$. The transport properties were also updated to use improved diffusion coefficients based on recent high level quantum calculations. The change in predictions to the *iso*-butanol experiments caused by the substitution of Burke et al. [57] model is within the various errors ranges associated with the simulation methods and the experimental measurements.

3.1.2. Unimolecular decomposition of *iso*-butanol

The pressure dependent network for unimolecular decomposition of *iso*-butanol was solved by Zhou et al. [58]. That study provides high level *ab-initio* analysis of three dominant decomposition pathways for *iso*-butanol.



Zhou et al. treated the radical forming channels with variable-reaction-coordinate transition state theory on a PES derived from CASPT2(2e,2o)/cc-pVDZ calculations while the dehydration channel was treated as simple transition state theory at CCSD(T) level of theory with Eckart tunneling [59] and 1-d hindered rotor corrections. They assumed $(\Delta E_d) \text{ cm}^{-1} = 200(T/298 \text{ K})^{0.85}$ when computing the $k(T,P)$. Recently the rate of dehydration of *iso*-butanol was measured by Rosado-Reyes et al. [60] and is within a factor of 2 of the *ab-initio* calculations by Zhou et al. The experiments also determined, in agreement with the calculations that this reaction system is not significantly pressure dependent up to 1250 K at 1.5 bar. The current model uses the calculated values by Zhou et al.

3.1.3. Hydrogen abstraction from *iso*-butanol by hydroxyl radical

Hydrogen atom abstractions by hydroxyl radical have been found to be an important set of reactions to predict auto ignition behavior of *n*-butanol. Though there have been various experimental [61–63] and theoretical studies [64,65] for *n*-butanol, only a few experimental studies for the overall rate of abstraction are available for *iso*-butanol [63,66]. Therefore we use analogy and theoretical calculations for *n*-butanol + OH rates to estimate the rate of hydrogen atom abstractions by hydroxyl radical from *iso*-butanol.

Abstraction from the alpha carbon is expected to be the most dominant channel and the rate is approximated to be the same as that for the alpha carbon for *n*-butanol because of similar bond dissociation energies (BDEs) (alpha site for *n*-butanol is 95.5 kcal/mol while for *iso*-butanol it is 95.9 kcal/mol at CBS-QB3) as shown in Fig. 2. This study employs the rate constant from Zádor and Miller [65] for abstraction from the alpha site instead of the values recommended by Zhou et al. [64] because of better agreement with low temperature experimental data, as discussed by Sarathy et al. [22]. The hydrogen abstraction from the beta carbon in *iso*-butanol is from a tertiary carbon site and so no good analogy exists from *n*-butanol (due to lack of any tertiary carbon). The rate constant is approximated as the beta carbon abstraction rate for *n*-butanol as calculated by Zhou et al. [64] with activation energy corrected by Evans–Polyani type of correlation [67].

$$E_{\text{iso-butanol}} = E_{\text{n-butanol}} + 0.4 \times (\text{BDE}_{\text{n-butanol}} - \text{BDE}_{\text{iso-butanol}}) \quad (1.5)$$

The gamma site in *iso*-butanol is the abstraction from primary carbon site and is estimated by the structure activity relationship given by Kwok et al. [68] and is within 20% of hydrogen abstraction by hydroxyl radical from primary site of *n*-butane [69]. Abstraction

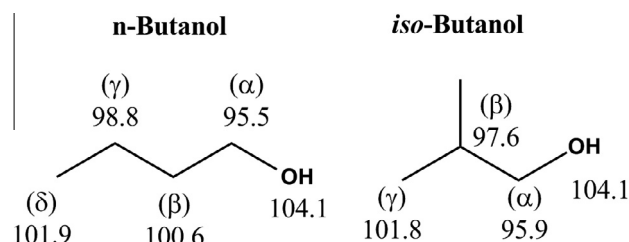


Fig. 2. Comparison of C–H bond energies (kcal/mol) for *n*-butanol and *iso*-butanol calculated at CBS-QB3 level of theory [22].

from the hydroxyl moiety is the slowest channel and is typically unimportant for *iso*-butanol; this study employs the rate constant as calculated by Zhou et al. [64].

All rate coefficients for hydrogen atom abstraction from *iso*-butanol by OH are summarized in Table 2. The estimate is compared to the experimentally measured rate constant from Pang et al. [63] and is within 20% over the temperature range of 900–1150 K. It is also within a factor of two for the low temperature experimental data from Mellouki et al. [66]. Since the experiments only give the total rate of abstraction from all the sites a high level calculation or experimental branching ratio measurement would be helpful in verifying these estimates. Currently we estimate an uncertainty in estimated individual rates to be factor of 2–5.

3.2. Quantum calculation and Statistical mechanics

The sensitivity analysis (discussed in detail in Section 4) over different experiments identified key species and reactions. For these important species the enthalpy of formation, the entropy at 298 K and the heat capacity at the following temperatures 300, 400, 500, 600, 800, 1000 and 1500 K were calculated. These parameters were then fit to a NASA polynomial form for use in the CHEMKIN input file. CBS-QB3 calculations [70–72] were done using the Gaussian 03 suite [73] to obtain electronic energies. The vibrational frequencies are calculated using B3LYP/CBSB7 method and are uniformly scaled by factor of 0.99 [74]. The RRHO with 1-d hindered rotor correction for every heavy atom-heavy atom dihedral

was implemented. The 1-d hindered rotor potentials were done at B3LYP/CBSB7 using a step size of 10° and relaxing all other degrees of freedom. The resulting energies were fit to a Fourier series as a function of dihedral angle ϕ . The reduced moment of inertia of the equilibrium geometry was computed and resulting one-dimensional Schrodinger equation was solved for the lowest 200 energy levels. These energies were used to compute the partition function and desired thermodynamic quantities. For the heats of formation, the Bond Additivity Corrections (BAC) to CBS-QB3 as proposed by Petersson et al. [75] were applied. The reported (forward) rate coefficients are from ab-initio calculations. The BAC does not affect these rate coefficients but will affect the reverse rates used in the model due to change in equilibrium constant K_{eq} .

The high pressure limit rate coefficients were calculated using the transition state theory as given in the following equation

$$k = \kappa(T) \frac{k_b T}{h C^\circ} \exp \left(-\frac{G^{TS} - \sum_{\text{react}} G^i}{RT} \right) \quad (1.6)$$

where k is the reaction rate coefficient, k_b is Boltzmann's constant, T is temperature, h is Planck's constant, R is the universal gas constant, C° is the standard state concentration ($C^\circ = P^\circ/RT$), G^i is the free energy of species i at concentration C° . $\kappa(T)$ is the asymmetric Eckart tunneling [59] correction and was calculated using the formalism presented by Johnston and Heicklen [76] with their eq. 15 corrected by Garrett and Truhlar [77]. All these calculations are performed using an in house code CANTHERM [78]. Table 3 gives the list of species and their calculated thermochemistry and Table 4

Table 2
Estimated modified Arrhenius rate coefficient parameters for H-atom abstraction reactions by OH radicals (see text).

	Reaction	A (cm ³ /mol s)	n	E_a (kcal/mol)
1	iBuOH + OH = (CH ₃) ₂ CHCHOH + H ₂ O	3.61E+03	2.89	−2.29
2	iBuOH + OH = (CH ₃) ₂ CCH ₂ OH + H ₂ O	1.54E+00	3.7	−4.94
3	iBuOH + OH = CH ₂ CH(CH ₃)CH ₂ OH + H ₂ O	5.4E+06	2.0	0.51
4	iBuOH + OH = (CH ₃) ₂ CHCH ₂ O + H ₂ O	5.88E+02	2.82	−0.58

Table 3
Species thermochemistry properties calculated using CBS-QB3 RRHO with 1-D hindered rotor corrections^a.

	$H_{f,298}$	S_{298}	C_{p300}	C_{p400}	C_{p500}	C_{p600}	C_{p800}	C_{p1000}	C_{p1500}
iBuOH	−68.34	83.38	27.21	33.74	39.68	44.72	52.64	58.55	67.90
(CH ₃) ₂ CHCHOH	−24.24	84.93	27.12	33.12	38.42	42.88	49.87	55.11	63.43
(CH ₃) ₂ CCH ₂ OH	−22.94	84.64	27.11	32.27	37.18	41.54	48.70	54.18	62.91
CH ₂ CH(CH ₃)CH ₂ OH	−18.28	86.11	28.03	33.68	38.81	43.16	49.99	55.12	63.33
(CH ₃) ₂ CHCH ₂ O	−15.78	80.19	26.32	32.81	38.45	43.15	50.50	55.97	64.52

^a Enthalpy has units of kcal mol^{−1}; entropy and heat capacity have units of cal mol^{−1} K^{−1}.

Table 4
Modified Arrhenius rate coefficients parameters, computed using conventional transition state theory at CBS-QB3 for important reactions involving *iso*-butanol.^a

	Reaction	A (cm ³ /mol s)	n	E_a (kcal/mol)
1	iBuOH + H = (CH ₃) ₂ CHCHOH + H ₂	1.09E+07	1.59	3.35
2	iBuOH + H = (CH ₃) ₂ CCH ₂ OH + H ₂	1.74E+07	1.48	3.44
3	iBuOH + H = CH ₂ CH(CH ₃)CH ₂ OH + H ₂	2.70E+07	1.76	7.45
4	iBuOH + H = (CH ₃) ₂ CHCH ₂ O + H ₂	4.05E+04	2.38	9.34
5	iBuOH + CH ₃ = (CH ₃) ₂ CHCHOH + CH ₄	1.87E+00	3.50	6.0
6	iBuOH + CH ₃ = (CH ₃) ₂ CCH ₂ OH + CH ₄	4.14E+02	2.87	4.89
7	iBuOH + CH ₃ = CH ₂ CH(CH ₃)CH ₂ OH + CH ₄	1.61E+00	3.59	7.7
8	iBuOH + CH ₃ = (CH ₃) ₂ CHCH ₂ O + CH ₄	2.32E+00	3.49	6.0
9	iBuOH + C ₂ H ₃ = (CH ₃) ₂ CHCHOH + C ₂ H ₄	5.19E−02	3.9	0.86
10	iBuOH + C ₂ H ₃ = (CH ₃) ₂ CCH ₂ OH + C ₂ H ₄	2.69E−02	3.9	0.68
11	iBuOH + C ₂ H ₃ = CH ₂ CH(CH ₃)CH ₂ OH + C ₂ H ₄	1.10E−03	4.55	3.5
12	(CH ₃) ₂ CHCHOH + H ₂ O ₂ = iBuOH + HO ₂ ^b	5.69E+01	3.04	1.2
13	(CH ₃) ₂ CCH ₂ OH + H ₂ O ₂ = iBuOH + HO ₂ ^b	2.91E+00	3.31	1.4
14	CH ₂ CH(CH ₃)CH ₂ OH + H ₂ O ₂ = iBuOH + HO ₂ ^b	3.01E+00	3.28	1.0

^a The rate coefficient expression is $k = A(T/K)^n \exp(-E_a/RT)$.

^b w/o 1-d hindered rotors correction.

reports the list of reactions and their fitted modified Arrhenius parameters. The CBS-QB3 method has an uncertainty of 2–3 kcal/mol in the barrier height [79] and the 1-d hindered rotor approximation has an uncertainty of factor of ~ 2 (only when there is no strong coupling between the torsions and other degrees of freedom) in the A factor [80]. In absence of any hindered rotor model (i.e. harmonic oscillator assumption) the uncertainty in the A factor is greater than an order of magnitude [81]. It is important to note for *iso*-butanol + HO₂ the rates reported in Table 4 are in the exothermic direction and the important endothermic direction is additionally sensitive to uncertainties in thermochemistry of *iso*-butanol and the corresponding radical.

4. Results

4.1. Pyrolysis experiment

Figure 3 reports the conversion of *iso*-butanol at diluted and undiluted conditions and shows that the predicted *iso*-butanol conversion versus the average process gas temperature properly matches the experimental measurements. Slightly higher conversions are observed experimentally than simulated, but deviations are less than the uncertainty on the measured conversions. The temperature range that has been studied in the pyrolysis experiments covers almost the complete conversion range for *iso*-butanol, and hence, the average process gas temperature in the hot zone of the reactor is a good measure for the cracking severity for the present set of pyrolysis experiments.

In the Supporting information the results obtained with the model developed by Sarathy et al. and the results obtained with the model of Frassoldati et al. are shown for the *iso*-butanol conversion, and the yields of ethene, formaldehyde and benzene. The model of Sarathy and coworkers does not include any reactions for aromatic formation nor important soot precursors such as 1,3-cyclopentadiene. The model of Frassoldati et al. predicts the conversion and the yields of the main products correctly although many irreversible, lumped reactions are used by these authors.

Figure 4 reports the yields of some of the major species obtained from the pyrolysis of *iso*-butanol. Methane, CO, alkenes as well as major oxygenated species such as formaldehyde and water are properly predicted by the kinetic model. Enols cannot be detected by the analytical techniques employed in this work because surface-catalyzed reactions rapidly convert enols into the

corresponding aldehydes (or ketones) [82]. In order to compare the measured and simulated concentrations the enol and aldehyde concentrations from the simulations have been summed and these have been matched against the measured values, see for example Fig. 4f. The presented mechanism simulates the trend for propanal well, although the maximum in the yield is underestimated by the kinetic model.

A good agreement is also observed for the yields of the minor species such as hydrogen, ethane, CO₂, acetone, 1,3-butadiene, benzene and toluene, reported in Fig. 5. The effect of the dilution is most pronounced for ethane in line with earlier work [83]. The yield of 1,3-cyclopentadiene is correctly estimated using the present model. The latter can be important for the simulation of soot formation, where the cyclopentadienyl radical plays an essential role [84–86].

To gain further insight in the reaction mechanism and the role of certain kinetic parameters, a rate of production analysis is carried out for the pyrolysis experiments. Figure 6 shows the schematic picture of the main *iso*-butanol decomposition pathways. Hydrogen abstraction reactions by methyl and hydrogen are predominantly responsible for the conversion of *iso*-butanol producing at the same time methane and hydrogen in line with the observations for the other butanol isomers. For *iso*-butanol the beta scission of the 3-hydroxy-2-methylpropyl radical and the *iso*-propyl radical formed after initiation is the dominant path toward propene. A smaller contribution to propene comes from the *iso*-butyl radical that is formed by a hydrogen addition reaction to isobutene. Most of the *iso*-butene is formed directly by the dehydration of *iso*-butanol. The radical pathway starting from the 2-hydroxy-1,1-dimethylethyl radical leads to only one third of the isobutene formation. The main route towards ethene is the beta scission of the *n*-propyl radical that originates almost entirely from propene. The direct dehydration of *iso*-butanol and hydrogen abstractions by OH radicals explains the significant amounts of water measured. The formation of formaldehyde and methanol is related to the formation of the CH₂OH radical from *iso*-butanol and the decomposition of ketene that is formed from acetone. Methanol and formaldehyde are the main intermediates towards CO and finally CO₂ in line with earlier work [26,30].

Aromatics are partly formed by a series of pressure dependent addition reactions such as the addition of the vinyl radical to 1,3-butadiene [87] leading to benzene. An equally important path is the pressure dependent recombination of two allyl radicals, which form the 1,5-hexadiene molecule in our mechanism. Hydrogen abstraction reactions by radicals such as the methyl or hydrogen radical give rise to the 2,5-hexadienyl radical. The endo cyclization reaction gives a six membered ring, while the exo cyclization gives a five membered ring. The former will result in benzene, while the latter will result in the formation of 1,3-cyclopentadiene after a beta scission reaction of the methyl-cyclopentenyl radical.

The addition and recombination reactions of the resonantly stabilized *iso*-butenyl radicals seem to be particularly important for describing the formation of soot precursors such as 1,3-cyclopentadiene, benzene and toluene in *iso*-butanol pyrolysis, while for other fuels including renewable *n*-butanol they hardly seem to contribute. For example the second most important pathway to 1,3-cyclopentadiene starts from the recombination of the *iso*-butenyl radical with the ethyl radical or the addition of that *iso*-butenyl radical to ethylene. Again several hydrogen abstraction reactions and a cyclization reaction are needed to form a methyl-cyclopentyl radical which can form cyclopentene after a beta scission reaction. The latter will then further decompose to 1,3-cyclopentadiene via a combination of hydrogen abstractions and a C–H scission.

Another major part of the formation of aromatics under the present conditions according to the RMG generated mechanism originates from recombination reactions of two *iso*-butenyl

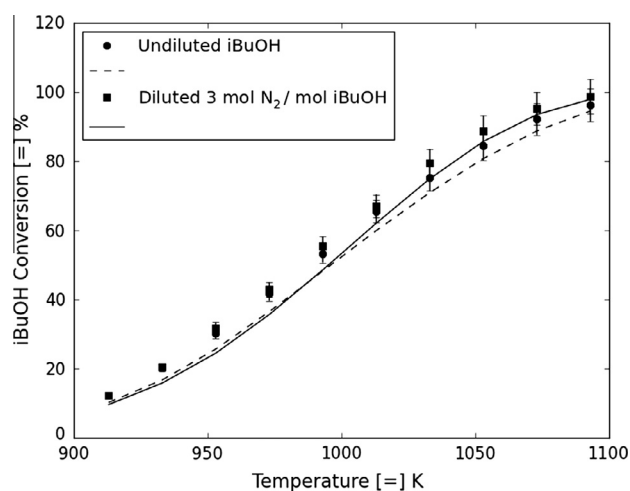


Fig. 3. *iso*-Butanol conversion as a function of the average pyrolysis reactor temperature (symbols – experimental: ● undiluted, ■ N₂ diluted: 3 mol N₂/mol *iso*-butanol, – simulated, -- undiluted and – diluted).

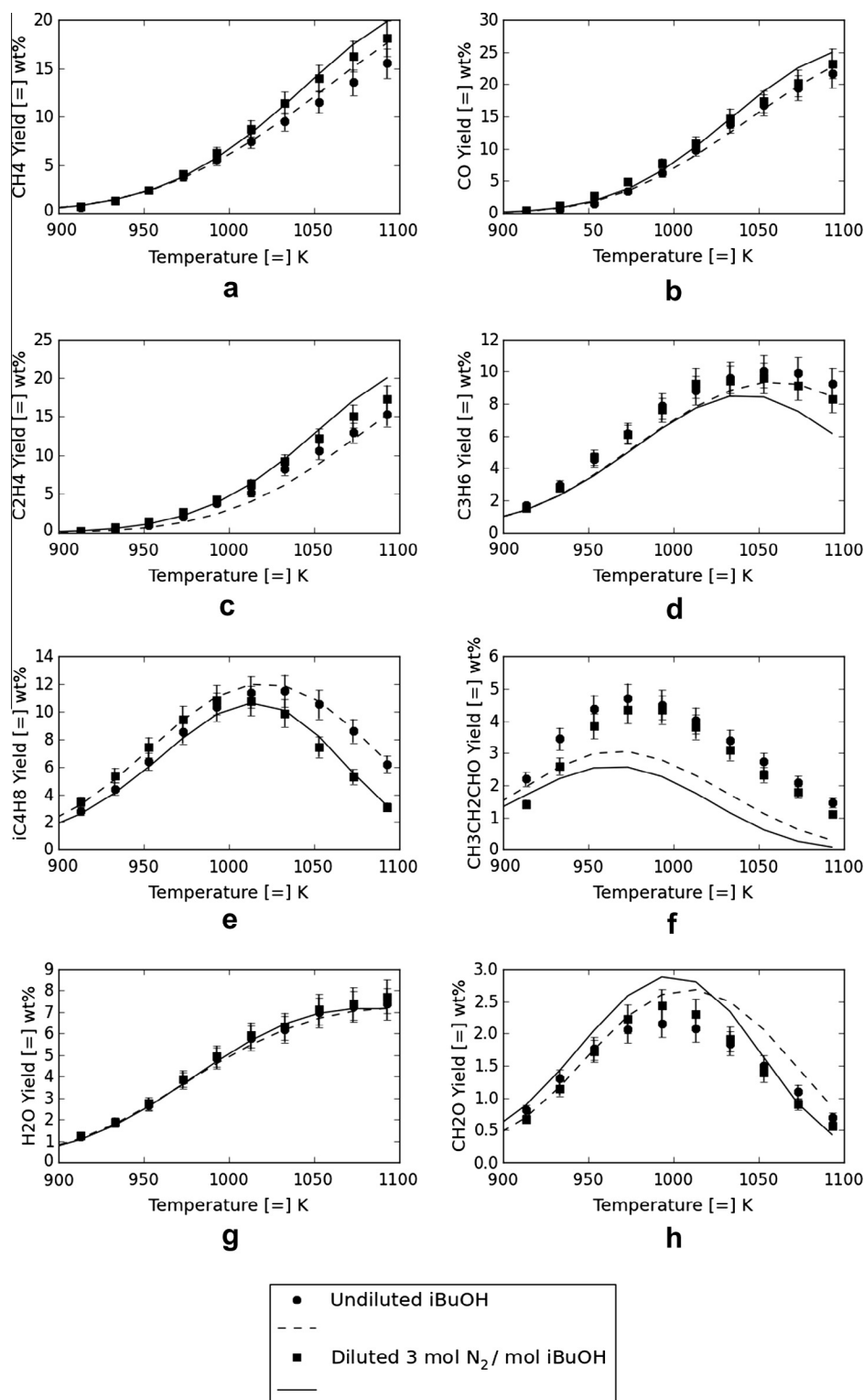


Fig. 4. Comparison of the experimental (symbols) and simulated (lines) yields of the major products at diluted (3 mol N₂/mol *iso*-butanol) and undiluted pyrolysis conditions. (a) Methane, (b) CO, (c) ethene, (d) propene, (e) isobutene, (f) propanal + prop-1-enol, (g) water, (h) formaldehyde (symbols – experimental, lines – model simulations).

radicals. This recombination leads to a C₈H₁₄ di-olefin, that after hydrogen abstraction followed by cyclization reactions and β -scission reactions can lead to methyl and di-methyl cyclohexadiene intermediates, as can be seen in Fig. 6. The presence of these cyclohexadiene isomers has also been observed experimentally; see for example the GC $\times$ GC chromatogram in Fig. 1. These cyclohexadiene isomers can rapidly form xylenes, toluene and even benzene.

The formation of 1,3-cyclopentadiene, well simulated as shown in Fig. 5d, is not only sensitive to the recombination of allyl with ethyl, but also to addition reactions. Naphthalene and indene are in our mechanism formed via the well-known addition and recombination pathways involving the cyclopentadienyl radical and 1,3-cyclopentadiene in line with work of the group of Violi et al. [88] and Ranzi et al. [89].

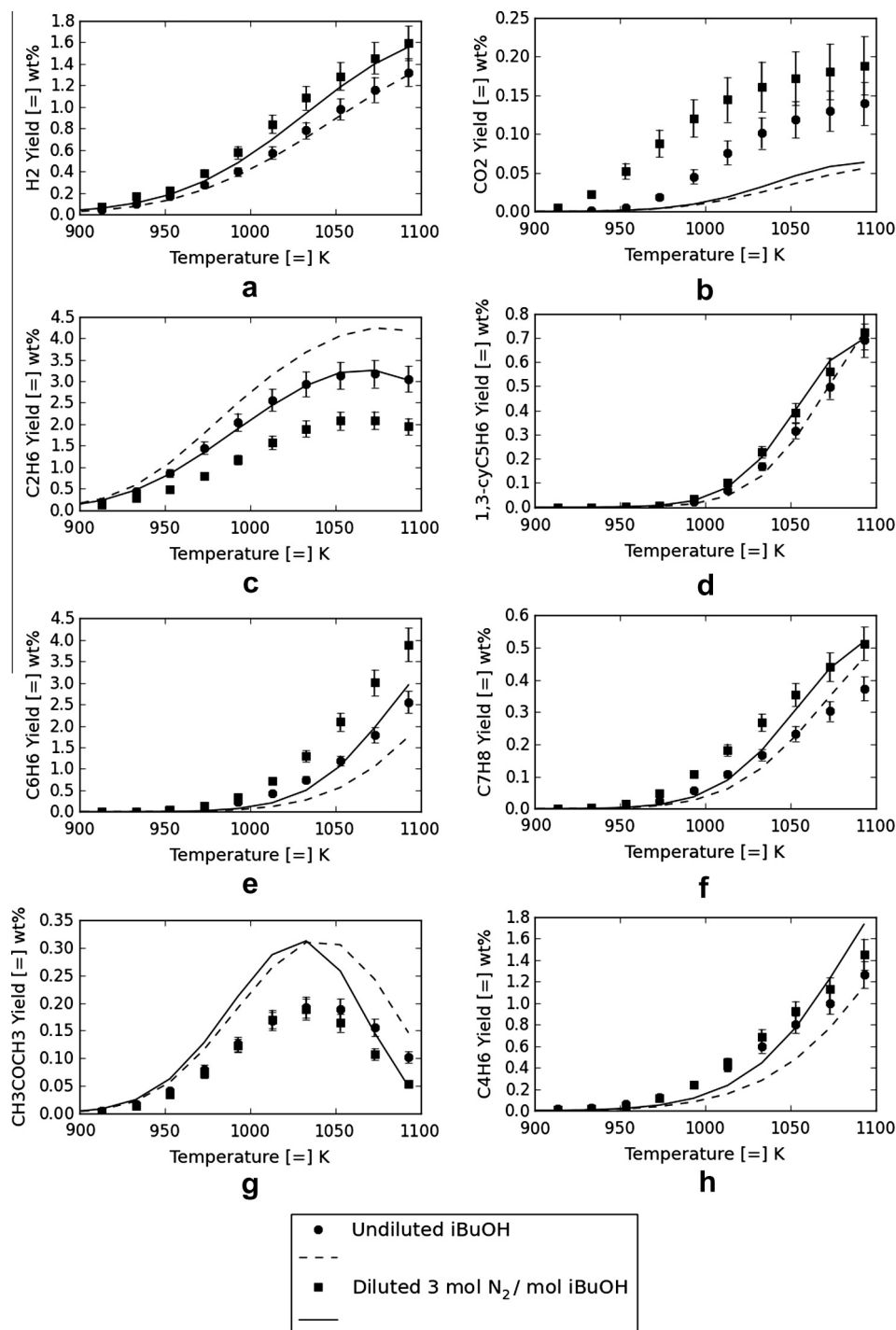


Fig. 5. Comparison of the experimental (symbols) and simulated (lines) yields of the minor products at diluted (3 mol N₂/mol *iso*-butanol) and undiluted conditions. (a) Hydrogen, (b) CO₂, (c) ethane, (d) 1,3-cyclopentadiene, (e) benzene, (f) toluene, (g) acetone, (h) 1,3-butadiene (symbols – experimental, lines – model simulations).

4.2. Doped methane flame

The model was validated against doped methane flame data of McEnally and Pfeifferle [15] using the flame solver discussed in Section 2.2.2. Figure 7 gives the simulated and experimental center-line *iso*-butanol profile. Experimentally it is seen that *iso*-butanol concentration drops to zero ~26 mm while the simulated profile reaches zero concentration ~29 mm, the 3 mm difference is within the uncertainty caused due to the experimental uncertainty of the measured gas temperature ~65 K (see Supplementary information

for figures showing the effect of assumed temperature profile on the axial position and peak value of the species profile).

4.2.1. Butanol flux analysis

Figure 8 gives the rate of production for the dopant *iso*-butanol. The main route of decomposition of *iso*-butanol is the C–C bond fission to give *iso*-propyl and hydroxymethyl radical. The *iso*-propyl radical undergoes β scission to give propene and H radical while hydroxymethyl radical undergoes β scission to give formaldehyde and H radical. The other significant pathways for *iso*-butanol

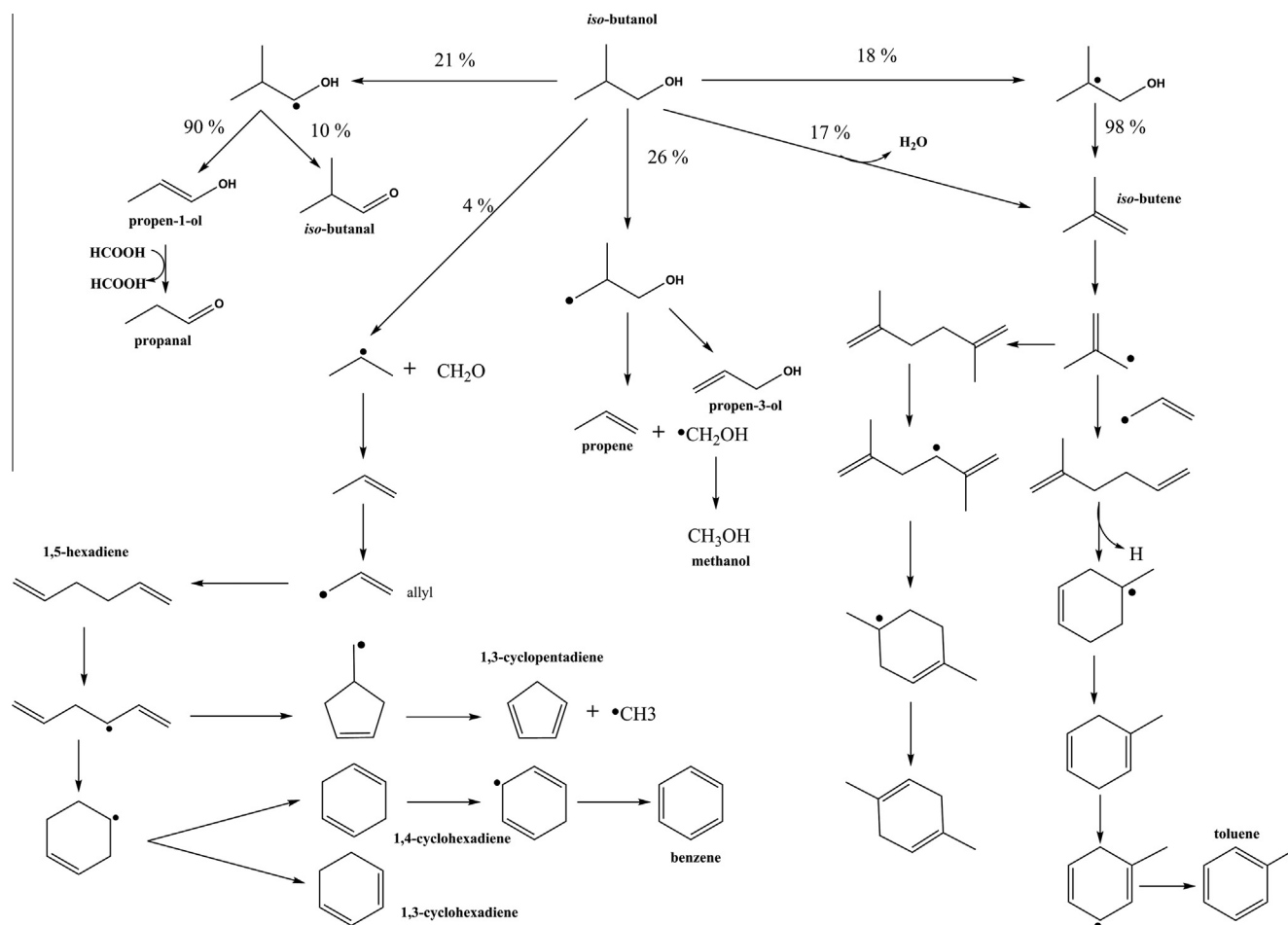


Fig. 6. Reaction pathway analysis for *iso*-butanol pyrolysis towards the main products performed at the maximum temperature along the reactor coil ($T_{\text{inl}} = 660$ K). The arrow percentages represent the branching fractions.

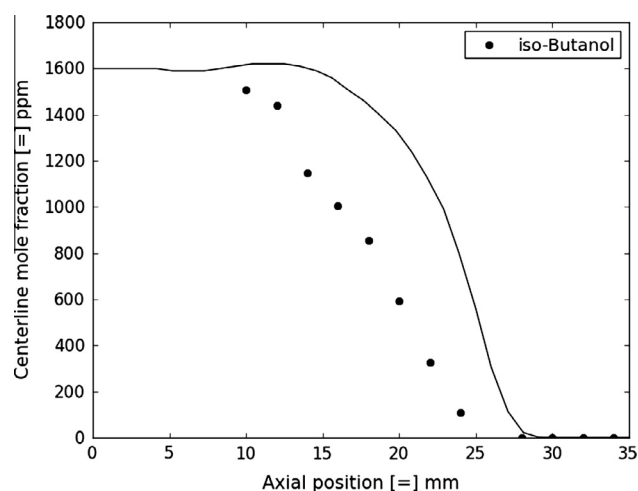


Fig. 7. Experimental (symbols) and simulated (line) centerline mole fraction profiles of *iso*-butanol in doped methane diffusion flame [15]. The offset along the X axis is mostly due to uncertainty in the assumed temperature profile, see Van Geem et al. [26].

decomposition are C–C bond fission to give 1-(hydroxymethyl)ethyl + methyl radical, hydrogen atom abstraction by methyl radical to give 2-hydroxy-1,1-dimethylethyl radical and dehydration of *iso*-butanol to *iso*-butene. The intermediates 1-(hydroxymethyl)ethyl and 2-hydroxy-1,1-dimethylethyl radicals undergo β scission to give propene and *iso*-butene respectively.

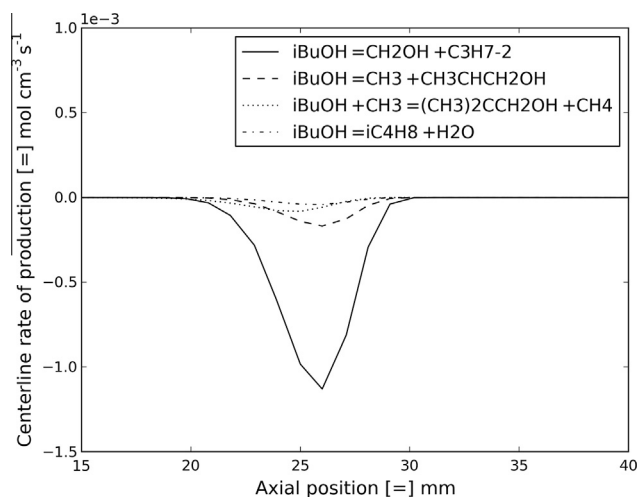


Fig. 8. Predicted centerline rate of production for *iso*-butanol in doped methane flame.

4.2.2. Butanol product centerline profiles and flux analysis

The centerline profiles of a few of the products of *iso*-butanol decomposition are given in Fig. 9; the species corresponding to each molecular weight are $M_w = 42$ (propene + ketene), $M_w = 44$ (acetaldehyde + vinyl alcohol), $M_w = 54$ (1,2-butadiene + 1,3-butadiene + 1-butyne + 2-butyne) and $M_w = 56$ (*iso*-butene). The model is able to predict profiles of all species well within the uncertainty

reported by McEnally and Pfefferle [15] (factor of 2 uncertainty in measured mole fractions) with exception of $M_w = 54$.

As seen from Fig. 9 the predicted peak concentration for propene ($M_w = 42$) is a factor of 1.75 greater than the experimental measured profile. Propene is the major product formed from *iso*-butanol decomposition as seen from flux analysis of *iso*-butanol discussed in previous section. Flux analysis shows the major consumption channel is a well skipping path of propene reacting with H atom to give ethene and methyl radical.

The flux analysis for acetaldehyde and vinyl alcohol are shown in Fig. 10a and b. The main route to vinyl alcohol is the well skipping reaction of ethene + OH, the rate coefficient for which is estimated by RMG. Acetaldehyde is formed primarily by tautomerization of vinyl alcohol and the well skipping reaction of ethene and hydroxyl radical to give acetaldehyde and H atom. The tautomerization rate coefficient was calculated by da Silva et al. [90]

($k = 8.59 \times 10^{11} T^{0.318} \exp[-55,900(\text{cal/mol})/RT]$) and the rate coefficient for the chemically-activated well skipping path is estimated using RMG. The primary decomposition path of acetaldehyde is C–C bond fission to give $\text{CH}_3 + \text{HCO}$ radical.

Flux analysis for *iso*-butene is given in Fig. 10c. The major routes for *iso*-butene formation are β scission of 2-hydroxy-1,1-dimethyl-ethyl radical to give *iso*-butene + OH and dehydration of *iso*-butanol to *iso*-butene. Unlike *sec* and *t*-butanol [26] where the butenes were formed almost exclusively by dehydration reaction, in *iso*-butanol the bond scission channel is the major channel in this temperature range (about 1300 K). The *iso*-butene is consumed primarily by CH_3 and H atom. The rate coefficients for the H atom abstraction from *iso*-butene come from CBS-QB3 calculations.

Species corresponding to $M_w = 54$ (1,2-butadiene + 1,3-butadiene + 1-butyne + 2-butyne) are under predicted by a factor of 3

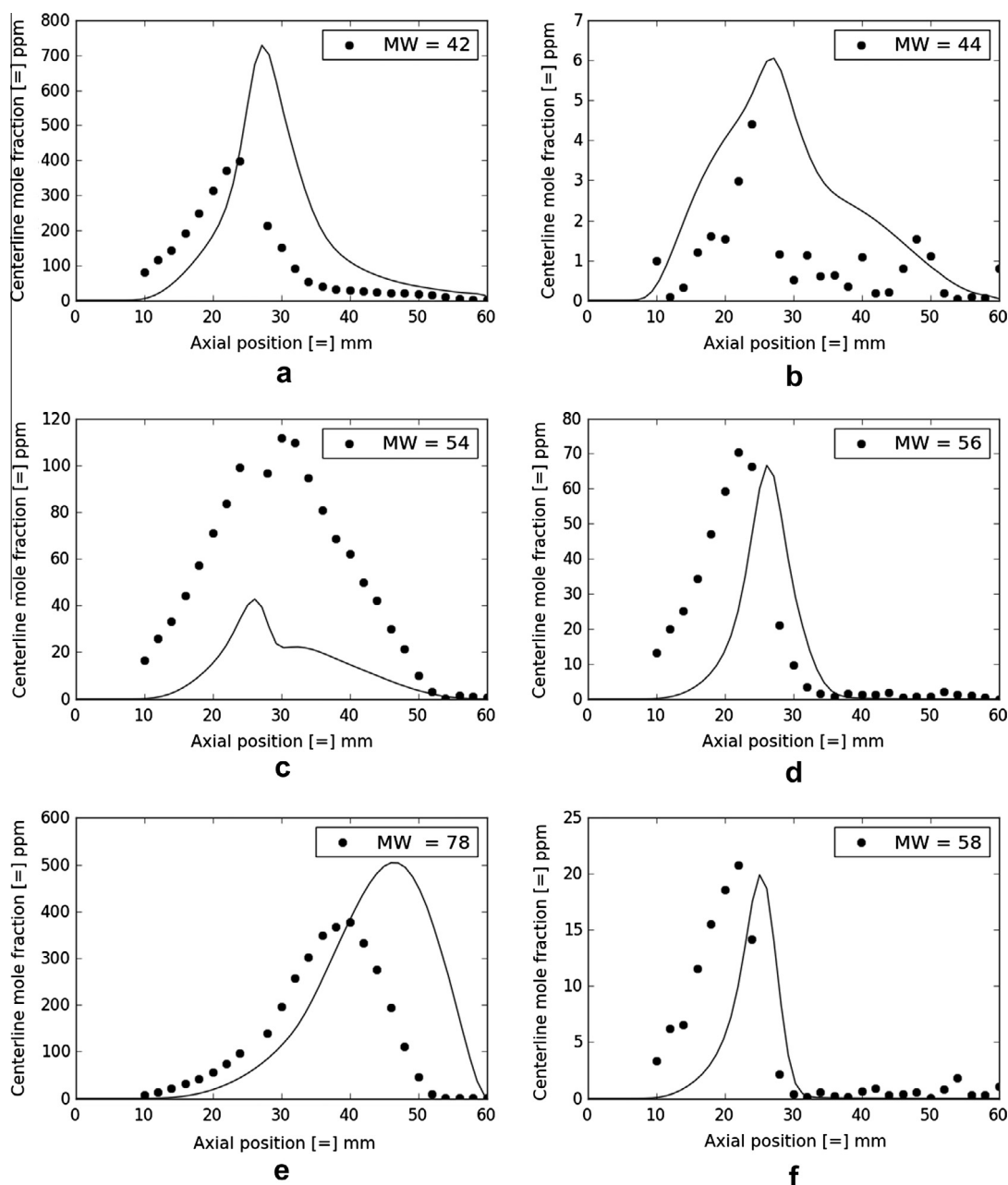


Fig. 9. Experimental (symbols) and simulated (line) centerline mole fraction of *iso*-butanol products in doped methane flame [15]. (a) $M_w = 42$ (mostly propene), (b) $M_w = 44$ (acetaldehyde + vinylalcohol), (c) $M_w = 54$ (1-butyne + 2-butyne + 1,3-butadiene + 1,2-butadiene), (d) $M_w = 56$ (mostly *iso*-butene), (e) $M_w = 78$ (mostly benzene) and (f) $M_w = 58$ (propanal + prop-1-enol + prop-2-enol).

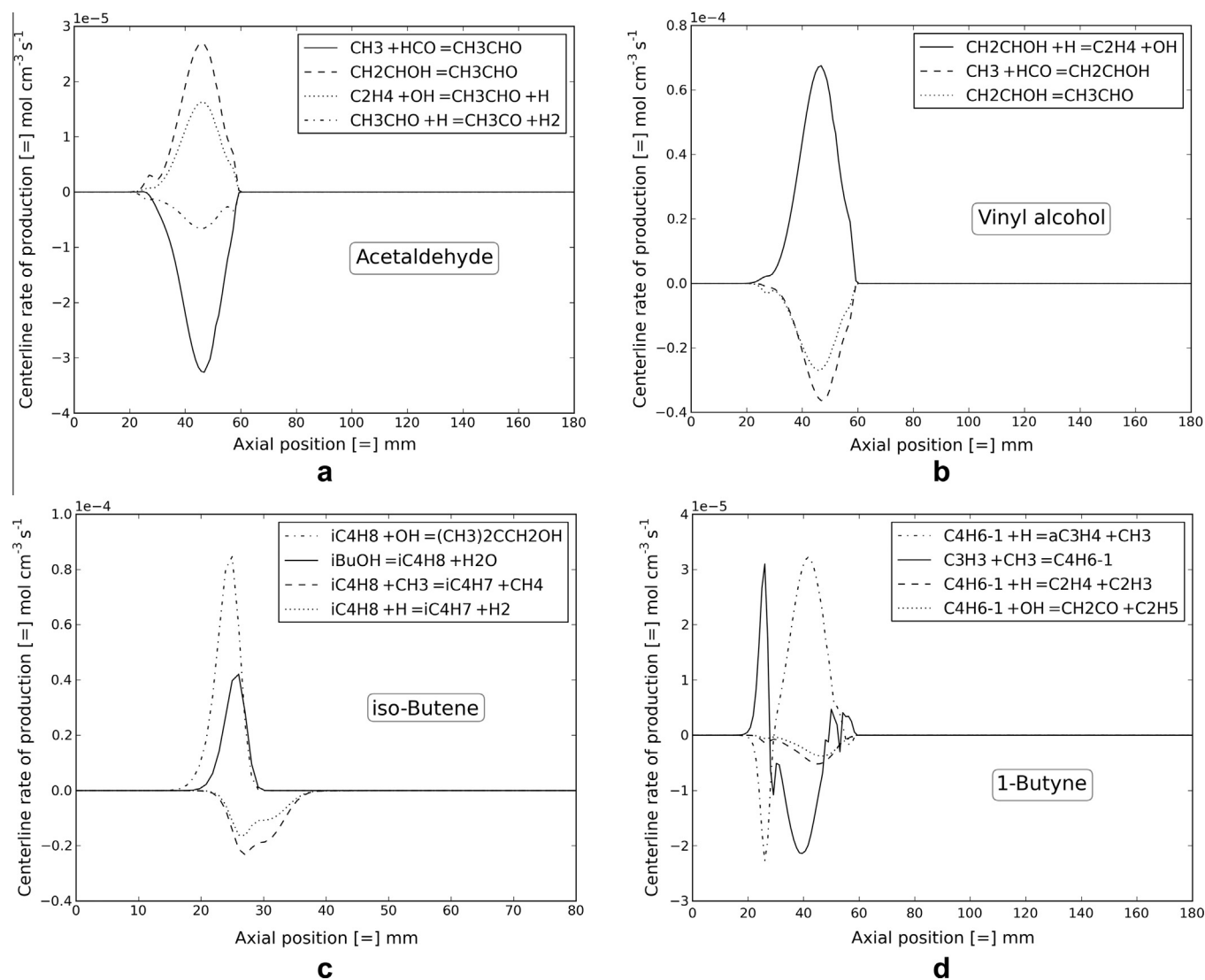


Fig. 10. Simulated centerline rate of production for (a) acetaldehyde, (b) vinyl alcohol, (c) *iso*-butene and (d) 1-butyne in *iso*-butanol doped methane diffusion flame. For (d) at low temperatures, propargyl + CH₃ form 1-butyne which is attacked by H to form allene. However at high temperatures the reactions run in reverse.

by the current mechanism. Figure 10d gives the flux diagram for 1-butyne (predicted to be the major species with $M_w = 54$). According to the model the major route for formation of 1-butyne is recombination of propargyl + CH₃ radicals to ~29 mm downstream from the burner. Further downstream the well skipping reaction of allene + CH₃ to give 1-butyne + H atom becomes dominant. The current mechanism save from the exceptions mentioned above predicts the centerline concentration of all other species including aromatics such as benzene (Fig. 9e) reasonably well. The complete comparison is in the [Supplementary information](#).

4.2.3. Sensitivity analysis to Lennard-Jones parameters

Normalized sensitivity coefficients were computed with respect to the Lennard-Jones σ and ϵ parameters for a few key species as shown in Fig. 11. The normalized sensitivity coefficient was calculated by numerical differentiation where the Lennard-Jones parameters were varied by $\pm 10\%$ from the nominal value. The centerline concentration profiles of most species were found to be sensitive to the collision diameter (σ parameter) of N₂ and CH₄ since they have the highest mole fraction in the flame. The value of these parameters were obtained from Burke et al. [57] and GRI-Mech 3.0 [55] mechanism respectively. The predicted species concentration

profiles are also sensitive to the collision diameter of the dopant *iso*-butanol. The value of σ_{iBuOH} used in this study is 5.667 Å and was computed using the group additivity scheme described in Ref. [25]. It is ~10% larger than the value in Ref. [91].

4.3. Jet-stirred reactor

The mechanism was also validated against the jet-stirred reactor data (JSR) of Togbé et al. [20]. For all the figures for this section “C₂H₄O” is the summation of acetaldehyde and its tautomer vinyl alcohol as these species were indistinguishable with the analytic equipment used in the experiment. Figure 12 gives the model prediction for major species for $\Phi = 0.5$. The comparisons of model prediction for other equivalence ratios are provided in [Supplementary information](#). As seen from Fig. 12 the *iso*-butanol consumption is not predicted accurately by the current model. The conversion is overpredicted for $T < 850$ K and underpredicted (by ~5–10%) as temperature is increased beyond 900 K. Figure 13 gives the flux analysis for the consumption of *iso*-butanol for all the experiments. For all three equivalence ratios, the three most dominant *iso*-butanol consumption reactions at temperatures < 1000 K are hydrogen abstractions by OH forming the 1-hydroxy-2-methylpropyl,

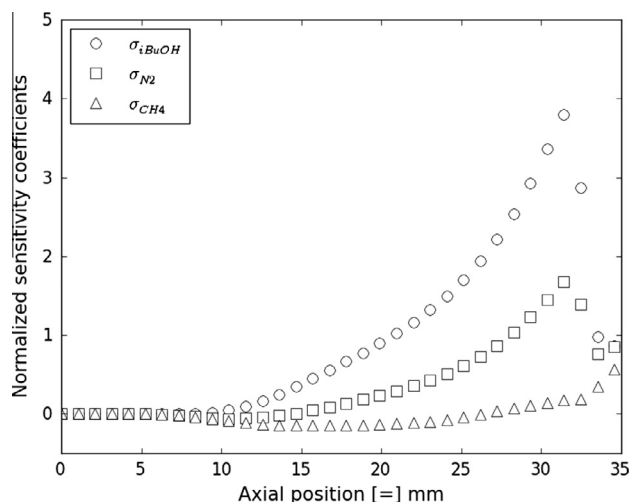


Fig. 11. Normalized sensitivity coefficients for the centerline *iso*-butanol concentration in the doped diffusion flame with respect to Lennard-Jones collision diameter σ for *iso*-butanol, nitrogen and methane. The *iso*-butanol concentration is near zero from 30 to 35 mm see Fig. 7.

2-hydroxy-1,1-dimethylethyl radical and 3-hydroxy-2-methylpropyl radicals. The rate coefficients for these reactions come from analogies to *n*-butanol as discussed in Section 3.1.3, hence have a high uncertainty associated with them and by adjusting these rate constants a better agreement between the predicted conversion can be achieved. However, for purpose of this study (testing the predictive power of the model) the rate constants are not adjusted. In the lean and stoichiometric cases for temperatures > 1100 K hydrogen abstraction by O atom forming 1-hydroxy-2-methylpropyl also becomes significant. In the rich case at high temperature the C–C bond fission to give *iso*-propyl and hydroxymethyl radical is the dominant pathway.

The model predictions for the concentration of CO, CO₂, H₂O, H₂, methane, ethene, formaldehyde, acetaldehyde + vinyl alcohol and acetylene are within a factor of two of the experimental data across the entire temperature range for all the equivalence ratios considered in the experiment.

iso-Butene concentration is under predicted for all the equivalence ratios as seen in Fig. 14, with the discrepancy increasing as we move from rich to lean conditions – at $\Phi = 0.5$ the peak concentration is under predicted by a factor of 6. Studies by Sarathy et al. [22] and Moss et al. [23] also under predict the peak concentration of *iso*-butene increasingly for rich to lean conditions. Figures 15 and 16 give the flux and sensitivity analysis respectively, for *iso*-butene formation for all the experiments. The predicted major pathway for formation of *iso*-butene for all three equivalence ratios over the complete range of JSR temperatures is the β -scission of the 2-hydroxy-1,1-dimethylethyl radical to give isobutene and hydroxyl radical. In the lean case the model predicts most of the 2-hydroxy-1,1-dimethylethyl adds molecular oxygen instead of undergoing β -scission to form *iso*-butene while for rich case most of the 2-hydroxy-1,1-dimethylethyl undergoes β -scission. Other pathways that contribute to production of *iso*-butene are the chemically activated pathway 2-methylprop-2-enol + H atom to *iso*-butene + OH and the β -scission of 1,1-dimethylpropyl radical (formed primarily by H addition to *iso*-pentene) to give *iso*-butene + CH₃. The rate expressions for the above reactions come from RMG estimates. The predominant decomposition pathway of *iso*-butene is abstraction of primary hydrogen by hydroxyl radical for lean and stoichiometric conditions while for the rich conditions the hydrogen abstraction by H atom is the dominant reaction especially at higher temperatures. The rate expression for abstraction by hydroxyl radical comes from Dooley et al. [92] while

abstraction by H atom is by quantum calculation as discussed in Section 3.2.

Figure 16a–c give the sensitivity analysis for *iso*-butene concentration at three different temperatures (800, 900 and 1100 K) for equivalence ratio $\Phi = 0.5$ (lean), $\Phi = 1$ and $\Phi = 4$ (rich). As seen from Fig. 16a the reactions which are most sensitive to *iso*-butene concentration are the hydrogen abstraction from the alpha and beta sites of *iso*-butanol by hydroxyl radical to form 1-hydroxy-2-methylpropyl and 2-hydroxy-1,1-dimethylpropyl radical. As explained in Section 3.1.3, at the time this model was generated there was very limited knowledge for these rate coefficients. Other reactions which are sensitive are the H abstraction from *iso*-butene by hydroxyl, and several small molecule reactions e.g. $H + O_2 = O + OH$, $HO_2 + O = O_2 + OH$, $HO_2 + OH = H_2O + O_2$, $HO_2 + CH_3 = OH + CH_3O$, $HO_2 + CH_3 = O_2 + CH_4$. The rate coefficients for reactions for the H₂/O₂ sub-mechanism come from the updated mechanism by Burke et al. [57] as stated previously, and the rate coefficients for HO₂ + CH₃ reactions come from the ab-initio quantum calculations by Jasper et al. [93].

At lower temperature the hydrogen abstraction from *iso*-butanol by HO₂ radical becomes important, the kinetics of which come from our CBS-QB3 RRHO quantum calculations (see Supplementary information). The system has strongly coupled internal rotors which are not treated accurately by current methods (uncertainty in the rate coefficients is an order of magnitude), so we recommend this reaction for future studies as discussed in Section 3.2. The sensitivity analysis for rich condition, $\Phi = 4$ gives similar results to that for lean conditions except that the hydrogen abstraction from the beta site of *iso*-butanol by hydroxyl radical has the highest sensitivity coefficient and hydrogen abstraction from *iso*-butene by hydrogen atom becomes important as temperature is increased.

A summary of the model predictions at different equivalence ratio is given in Table 5 where the peak experiment and simulation value for each species are compared for the entire dataset. Our model predicts the peak concentration of most species within a factor of 2 across all equivalence ratios, with the exception of the unsaturated aldehydes 2-propenal and 2-methylpropenal which are underpredicted significantly. The jet stirred reactor experiments measure ~20 ppm for both unsaturated aldehydes while the model predicts ~10 ppm for 2-methylpropenal and less than 2 ppm for 2-propenal for all equivalence ratios. Sensitivity analysis indicate the important reactions for formation of the unsaturated aldehydes are hydrogen abstractions by OH and HO₂ radicals from the corresponding saturated aldehydes. These rate coefficients have not been studied in literature and are RMG estimates with very high uncertainty (~ order of magnitude).

4.4. Opposed-flow diffusion flame

The mechanism was also tested against the opposed-flow diffusion flame data measured by Grana et al. [16] using the Opposed Flow Flame Module in Chemkin-Pro [31]. The experimental and modeling results are presented in Fig. 17.

The opposed flow diffusion flame was solved as an adiabatic system, with the experimental profile used as initial guess for the solver. The predicted temperature profile matches the experimental curve reasonably well: the maximum temperature is within 5% of the experimental values and well within the ± 200 K uncertainty reported by the experimentalists. The predicted temperature curve has a narrower reaction zone than the experimental one and the flame front leads by ~1.2 mm on the reactant side and lags by ~0.7 mm on the oxidizer side. The peak axial temperature position is within 0.5 mm of the experimental value. The model predicts the concentration profiles of all major species reasonably well (It must be noted that ± 200 K uncertainty in temperature

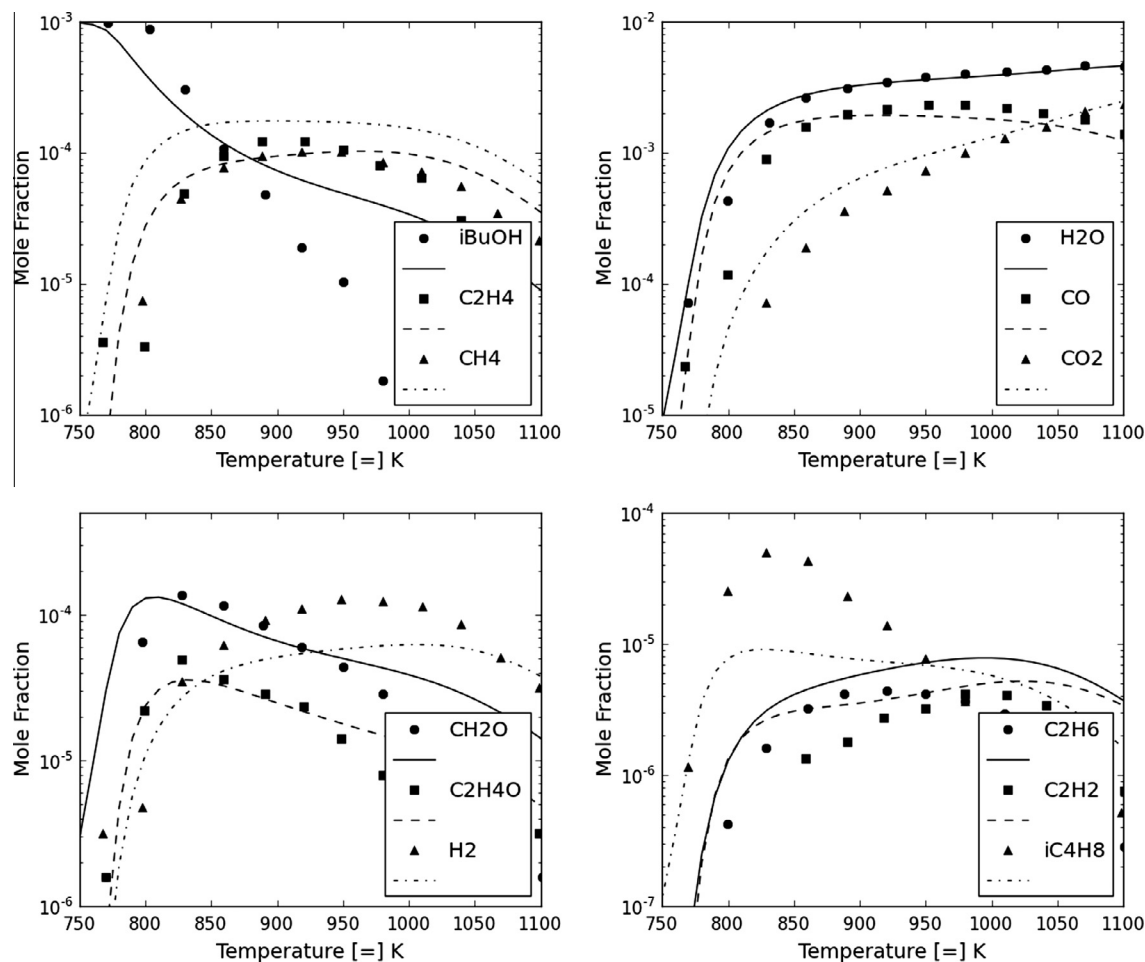


Fig. 12. Simulated (lines) and experimental (symbols) profiles of major species for jet stirred reactor experiment [20] for $\Phi = 0.5$ and $P = 10$ atm. (C_2H_4O is sum of acetaldehyde + vinyl alcohol).

represents a 20% uncertainty in the predicted peak concentration and ~ 0.5 mm uncertainty in the peak position of the intermediate species) as seen in Fig. 17.

The flux analysis of *iso*-butanol decomposition in this flame is shown in Fig. 18. The predominant decomposition path is the C–C bond fission of *iso*-butanol to give *iso*-propyl radical and hydroxymethyl radical with the maximum reaction rate occurring ~ 4.7 mm from the fuel port. Hydrogen abstraction by hydroxyl radical to give 1-hydroxy-2-methylpropyl and the C–C bond fission of *iso*-butanol to give methyl and 1-(hydroxymethyl)ethyl radical are other reaction pathways contributing significantly to *iso*-butanol consumption, but have a peak rate a factor of 6 smaller than the predominant decomposition path.

The peak values of CO, CO₂, H₂O and H₂ are predicted reasonably well. The axial position corresponding to peak position is slightly shifted towards the oxidizer side when compared to experimental profiles, but well within the experimental uncertainty. The model predicts the peak concentration profiles of ethene, acetylene, ethane, propene and *iso*-butene within 50% of the experimental value. Two species which the model currently does not predict well are formaldehyde and acetaldehyde overpredicted and underpredicted by a factor of 2 respectively. Figure 19 gives the flux analysis for major pathways leading to production of the aldehydes.

Figure 19a gives the flux analysis of formaldehyde. As seen from figure the model predicts that there is net production of CH₂O up to ~ 4.8 mm from fuel port, with peak production rate ~ 4.67 mm from fuel port. Beyond ~ 4.8 mm there is a net consumption of

CH₂O with maximum consumption rate at ~ 4.93 mm. The *iso*-butanol decomposes to give *iso*-propyl radical and hydroxymethyl radical, the hydroxymethyl radical then undergoes β -scission to give formaldehyde and H radical and this is the major path of production of formaldehyde up to ~ 5 mm from fuel port, beyond which the reaction of molecular oxygen with CH₂ radical is predicted to be the main formation route of formaldehyde. From 4.3 to 5.1 mm from the fuel port the main decomposition routes for formaldehyde are hydrogen abstraction by H atom and OH radical using the rate coefficients from GRI-Mech 3.0 [55]. The maximum consumption by OH occurs around 5.13 mm beyond which it rapidly decreases and becomes less important.

Figure 19b gives flux analysis predicted for acetaldehyde. Like formaldehyde there is net production of acetaldehyde near the fuel port, while at the oxidizer side (5–5.3 mm from fuel port) there is net consumption. The predominant decomposition route is predicted to be hydrogen abstraction from acetaldehyde by H atom to give acetyl radical with maximum rate ~ 5 mm from the fuel port. The production of acetaldehyde is from the well skipping reactions of propene + O atom and ethene + OH, predicted to have peak rates ~ 5 mm from the fuel port. Acetaldehyde is also formed by the well skipping reaction of H atom with vinyl alcohol and direct tautomerization of vinyl alcohol.

4.5. Laminar flame speeds

Laminar flame speeds are important properties of premixed flames and play an essential role in determining critical

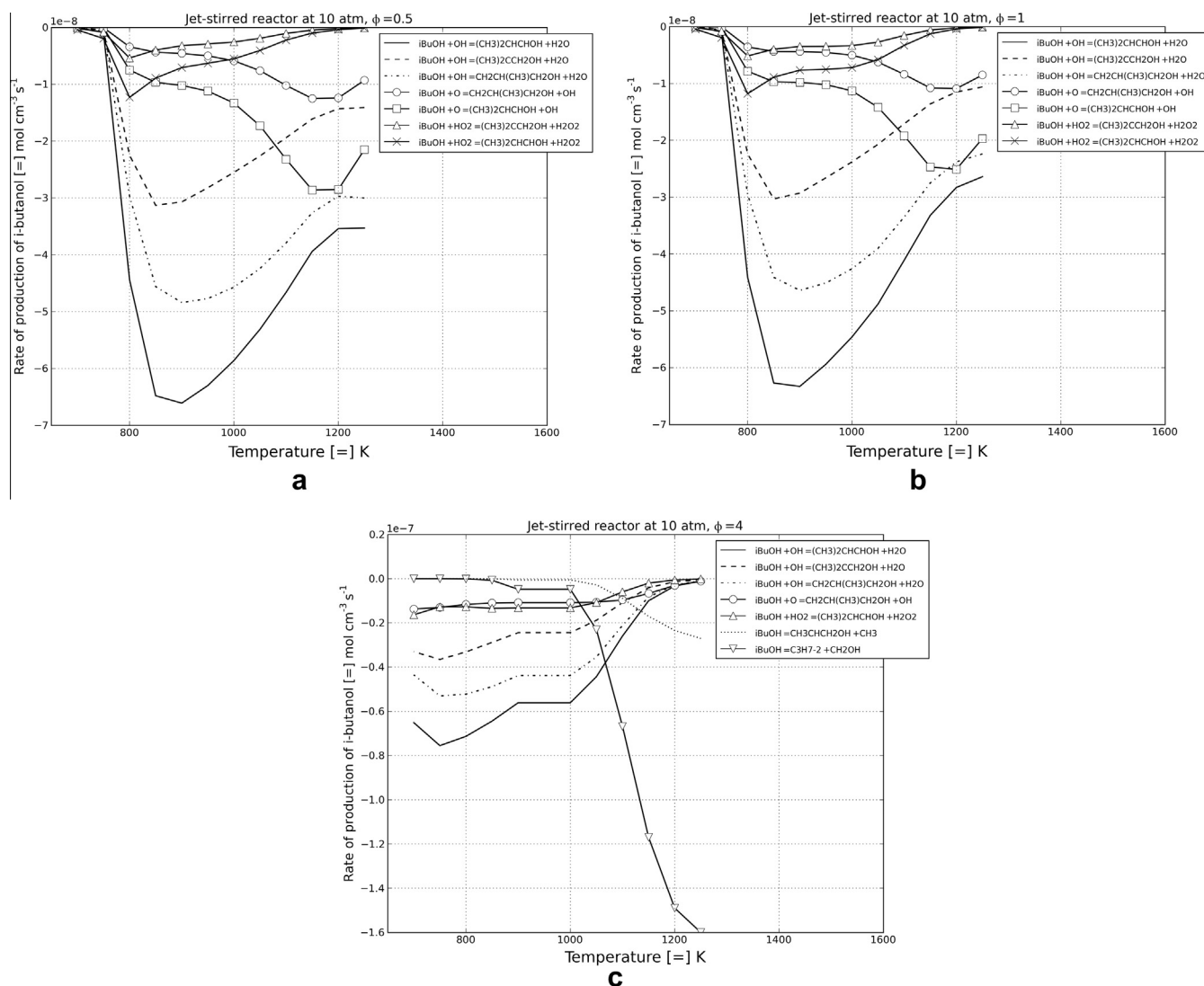


Fig. 13. Simulated rate of production for *iso*-butanol in the 10 atm jet stirred reactor [20] for equivalence ratios (a) 0.5, (b) 1 and (c) 4.

performance properties (i.e. knock, unburned hydrocarbon emissions) for spark ignition engines. The model is compared against available experimental data of *iso*-butanol/air laminar burning velocities measured by Veloo and Egolfopoulos [12] and Liu et al. [13]. The results were obtained in a range of equivalence ratios from 0.7 to 1.5 at 1 and 2 atm, but statistically at the same unburned gas temperature, i.e., there is no significant difference between the sets at 343 and 353 K because the uncertainty in temperature measurements was close to 20–30 K as reported by Liu et al. [13]. For the 1 atm study the model is able to accurately reproduce the peak flame speed at the correct stoichiometry (~ 1.1) and the flame speeds for the fuel lean case. However the model underpredicts the flame speeds for the rich case especially for experimental data by Liu et al. [13] as seen from Fig. 20. It must be noted here that the experimental data for laminar flame speed for *iso*-butanol at 1 atm by Veloo and Egolfopoulos [12] and Liu et al. [13] is in good agreement within the reported uncertainty for the lean and stoichiometric flames, but deviates beyond the reported uncertainty range for the fuel rich flames. Hence the uncertainty in the laminar flame speed measurement is quite significant on the rich side and it is possible that much of the discrepancy between predictions vs. data is due to experimental uncertainties. The experimental data for flame speeds at higher pressure is only

available from Liu et al. [13] and the model predictions are similar to 1 atm case, i.e. the experimental data is well predicted for the fuel lean flames but underpredicted for the rich flames.

The results of the sensitivity analysis for the rich flame at $\Phi = 1.4$, are reported in Fig. 21. The reactions with high sensitivity are the chain branching reaction $H + O_2 = O + OH$, thermal decomposition of methane ($CH_4 = CH_3 + H$) and ketene ($CH_2CO = HCCO + H$) to respective radical and H atom. Among these reactions the decomposition of ketene is most uncertain ($\sim$ order of magnitude) as the rate coefficient is calculated using a group contribution method implemented in RMG. The fuel rich flame also has notable sensitivity to fuel decomposition reactions like *iso*-butanol decomposing to *iso*-propyl radical and hydroxymethyl radical. Sensitivity analysis to heat of formation of species to laminar flame speed was also carried out (see Supporting material) and the thermochemistry for most of the sensitive species come from the active thermochemical tables (ATcT) [94] except for *iso*-butanol whose thermochemistry has been calculated using CBS-QB3 values with bond additivity corrections and has 1–2 kcal/mol uncertainty in the enthalpy.

The peak flame speed occurs close to stoichiometric condition and the sensitivity analysis at $\Phi = 1$ (Supporting information) reveals the thermal decomposition of formyl radical and the reaction of vinyl radical with molecular oxygen have high sensitivity.

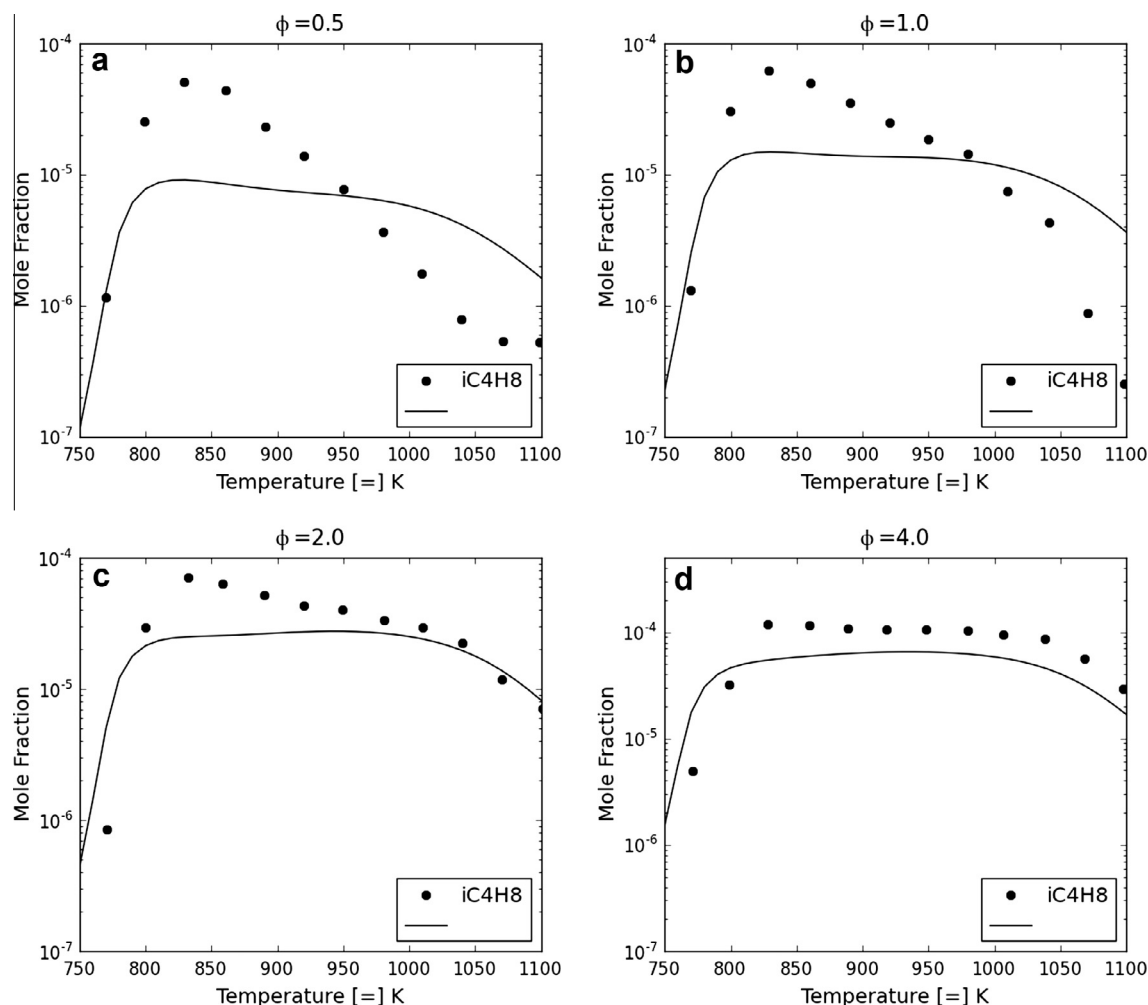


Fig. 14. Simulated (line) and experimental (symbols) profiles of *iso*-butene for jet stirred reactor experiment [20] for equivalence ratios (a) 0.5, (b) 1, (c) 2 and (d) 4.



The rate of decomposition of formyl radical has been recently reviewed by Li et al. [95] and the authors recommended the low pressure limit value of k_1 ($\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$) = $4.75 \times 10^{11} T^{0.66} \exp(-14,870 \text{ (cal/mol)}/RT)$ based on weighted least-square fitting to all experimental data. They also recommend a collision efficiency of 12 for water, since water is the major product of combustion this efficiency strongly affects the flame speed prediction. If collision efficiencies based on the ethanol reaction mechanism by Marinov [56] are used (collision efficiency of 5 for water) it underpredicts the peak flame speed by 3–4 cm/s.

Figure 22 presents a comparison between the most recent experimental [96,97] and theoretical [98–100] rate constants for reaction (2). The rates from RRKM calculations obtained by Mebel et al. [99] are close to the considered experimental results and are the values used in the present model. However, the Arrhenius expressions suggested by Bozzelli and Dean [98] and Marinov et al. [100], which are represented by the upper and lower dashed lines in the shadow region in Fig. 22, respectively are significantly slower than the experimental data. Except for the current mechanism and one for butanol oxidation proposed by Moss et al. [23], the expression suggested by Mebel et al. [99] has not been applied. For instance, Sarathy et al. [101] preferred to increase empirically the values of k_2 by a factor of 1.7 over the Mebel et al. [99] rate

constant, while Grana et al. [16] chose kinetic parameters ($k_2(\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}) = 1.0 \times 10^{12} \exp[-4000(\text{cal/mol})/RT]$) that are in great disagreement with the measured rates over almost the entire range of temperature in Fig. 22. Although the rate coefficient presented by Marinov et al. [100] for reaction (2) have not been applied for *iso*-butanol models, it was currently considered because it was widely accepted in combustion mechanisms for ethanol and primary reference fuels [56,102]. Moreover, it seems that Marinov et al. [100] has concluded a decrease in rate constants was needed as temperature is increased to have good predictions on ethanol flames. The authors feel that these reactions need special attention by kinetics community as even factor of 2 in the rate coefficients can noticeably affect the predicted flame speed.

The current mechanism for *iso*-butanol using the Mebel et al. [99] rate expression, gives predictions in accord with the experiments measured by Veloo and Egolfopoulos [12] and Liu et al. [13].

4.6. Ignition delay time

The last set of experiments for which the mechanism is validated are the autoignition delays for *iso*-butanol [17–19]. The experiments were modeled using the Closed Homogenous Batch reactor module in Chemkin-Pro. The reported ignition delay is the time at which half the maximum $[\text{OH}]$ is achieved. It corresponds well to the time at which maximum $[\text{C}_2\text{H}] * [\text{O}]$ is achieved and the time at which peak pressure rise rate is observed.

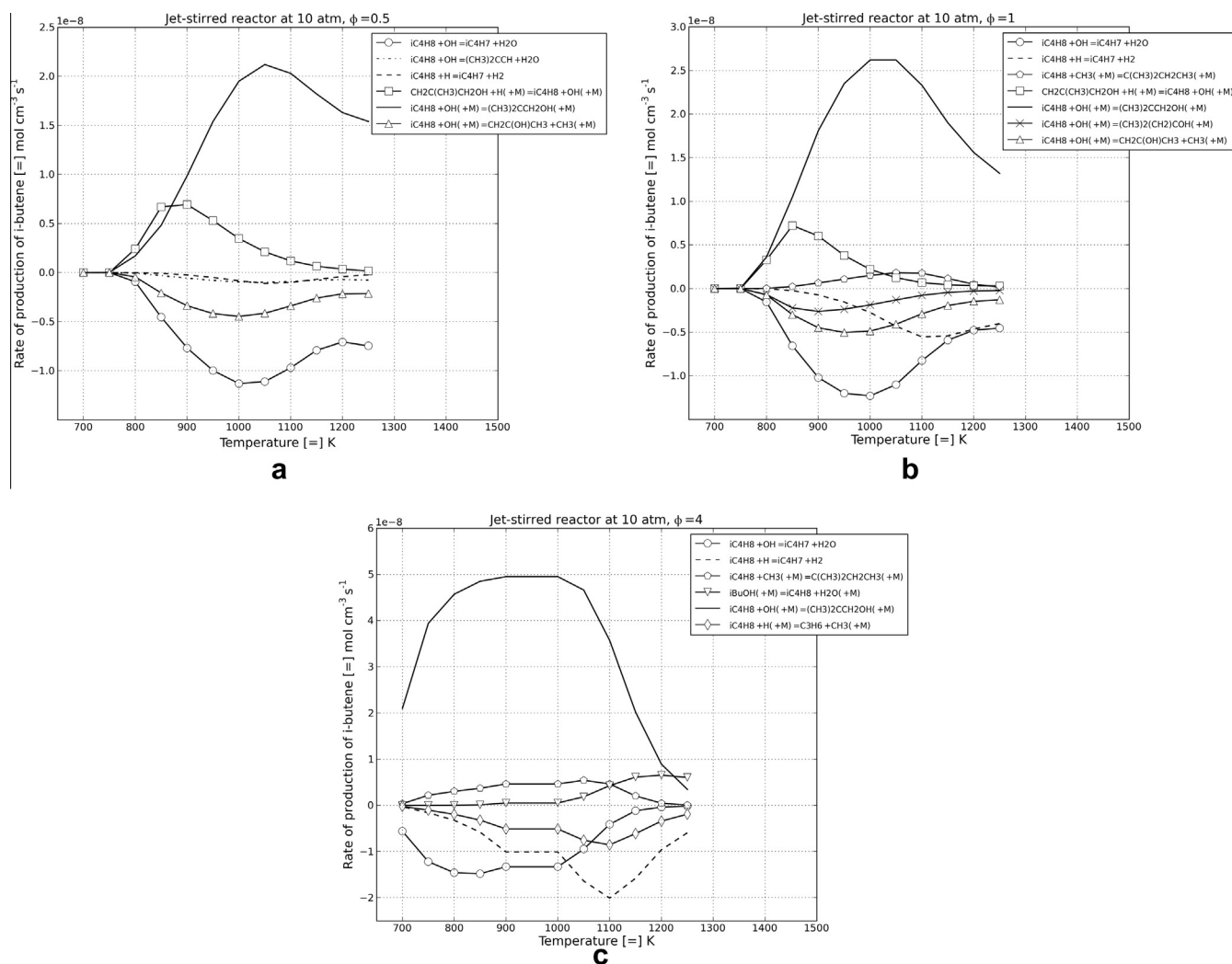


Fig. 15. Simulated rate of production for *iso*-butene in the 10 atm jet stirred reactor [20] for equivalence ratios (a) 0.5, (b) 1 and (c) 4.

4.6.1. High temperature shock tubes

4.6.1.1. Stanford shock tube experiments. Stranic et al. [17] measured autoignition delay times for *iso*-butanol behind reflected shock waves over a wide range of experimental conditions: 1250–1600 K, 1–43 atm and equivalence ratios of $\Phi = 1$ and 0.5. Overall as seen from Fig. 23 the model performs very well for the entire pressure range and at both the lean and stoichiometric conditions, the maximum deviation is within a factor of 2 for all conditions. Even with excellent agreement to the data, it is valuable to look at flux and sensitivity analysis to understand the major decomposition pathways for *iso*-butanol and sensitive parameters which are estimates to be looked into for future studies. Figure 24 gives the *iso*-butanol flux analysis for the simulation of *iso*-butanol autoignition, with 3% initial concentration of O_2 , $\Phi = 1$, $P = 1$ bar, and $T = 1400$ K. Initially the C–C bond fission to form *iso*-propyl radical and hydroxymethyl radical is the dominant decomposition route for *iso*-butanol. After 10 μ s there is enough buildup of OH radicals such that the flux through H abstraction of *iso*-butanol by hydroxyl radical becomes the dominant decomposition pathway leading to formation of 1-hydroxy-2-methylpropyl and 3-hydroxy-2-methylpropyl radical. The abstraction by hydroxyl radical to give 2-hydroxy-1,1-dimethylethyl radical is about an order of magnitude slower than the alpha and gamma sites at all times in the simulation, in part because there is only one beta hydrogen but six gamma hydrogens.

The shock tube ignition time is sensitive mainly to small molecule reactions, most of which have been discussed in previous sections. Additionally the C–C bond fission of *iso*-butanol to give *iso*-propyl radical plus hydroxymethyl radical is also sensitive. The rate coefficient for this has been recently calculated by Zhou et al. [58] and has been incorporated in the present model as discussed in Section 3.1.2.

4.6.2. Rapid compression machine experiments

Ignition delays were measured in a rapid compression machine (RCM) by Weber and Sung [18,19] for *iso*-butanol at $\phi = 1$, pressure of 15 and 30 atm with temperature range from 750 to 850 K. These studies provide an insight into the low temperature reactivity of *iso*-butanol. Figure 25 gives the comparison of the measured and predicted ignition delays for constant volume simulations (heat loss has not been incorporated). Incorporating the heat loss will increase the ignition delay by about a factor of 2.

The flux analysis (Fig. 26) for *iso*-butanol at $\phi = 1$ and 750 K reveal that the hydrogen atom abstraction by hydroxyl radical and O atom are the major routes of consumption beyond 0.2 sec. Earlier most *iso*-butanol consumption is by hydroperoxy radicals. As discussed in previous sections currently there is a significant uncertainty associated with both the *iso*-butanol + OH rate coefficient and *iso*-butanol + HO_2 rate coefficients (current study uses CBS-QB3 RRHO rate coefficients). It is therefore clear that given

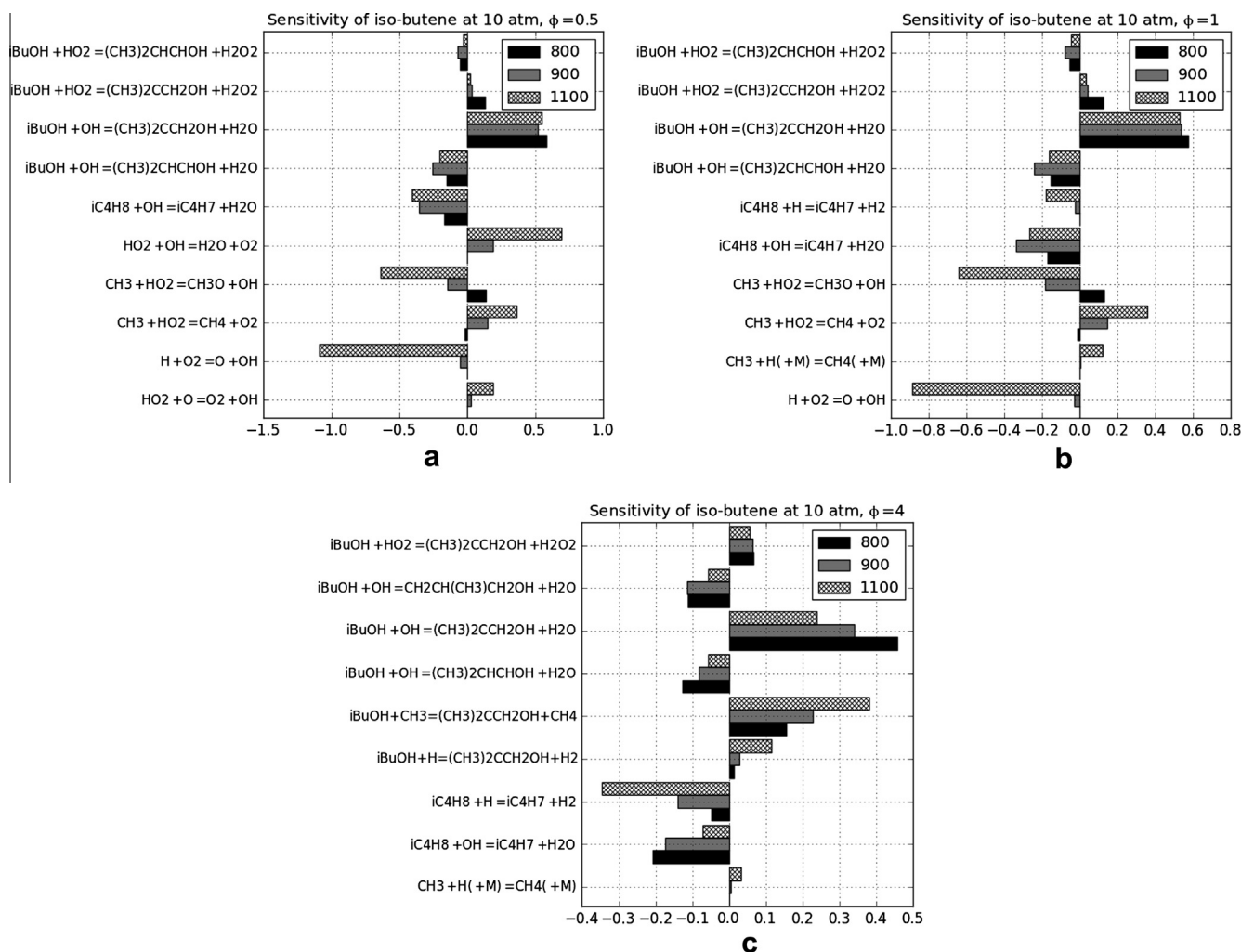


Fig. 16. Sensitivity analysis for *iso*-butene production in jet stirred reactor [20] at 10 atm as function of temperature (800–1100 K) and equivalence ratios (a) 0.5, (b) 1 and (c) 4.

the model's strong dependence on these reactions that more accurate values are needed either from theory or from experiments.

Table 5

Comparison of measured and predicted peak product concentration, rounded to the nearest ppm, in the 10 atm jet-stirred reactor of [20]. Italicized numbers represent measured values and bold numbers are the predicted values using the new mechanism.

	$\Phi = 0.5$	$\Phi = 1.0$	$\Phi = 2.0$
Formaldehyde	134	126	124
	132	135	139
Methane	103	210	622
	175	256	467
Acetaldehyde + vinyl alcohol	47	37	39
	36	33	29
Ethene	124	199	418
	103	187	302
Propene	203	219	228
	180	172	159
Acetone	65	46	35
	25	33	35
Propanal ^a	22	30	37
	16.7	22.5	33
<i>i</i> -Butene	50	62	70
	9	15	28
2-Methylpropanal ^a	138	137	143
	158	152	132

^a The -al is the sum of the aldehyde (-al) + alcohol (-ol) tautomer form.

5. Conclusion

A detailed, robust reaction mechanism for *iso*-butanol has been constructed using an automated, open-source software package, Reaction Mechanism Generator (RMG). The model has been validated against multiple types of experiments – flames, jet-stirred reactors, shock tubes – including new speciation data for pyrolysis of *iso*-butanol in a plug flow reactor. The validation targets selected cover a wide range of reaction conditions – diffusion and premixed, rich and lean, over a pressure range of 1–45 atm and a temperature range of 850–1800 K.

In general good agreement is seen for the prediction of the product distribution under pyrolysis conditions including soot precursors like 1,3-cyclopentadiene, benzene and toluene. Unlike for classical fuels, where addition and recombination reactions of vinyl, allyl and propargyl radicals are the dominant pathways towards soot formation, the most important pathways for aromatic formation in *iso*-butanol pyrolysis involve pressure dependent reactions of the resonantly stabilized *iso*-butenyl radical. Reducing the uncertainty on the rate coefficients for addition and recombination reactions involving the *iso*-butenyl radical is essential to completely understand how soot is formed for a fuel containing *iso*-butanol. This finding has important implications for the engine modeling community which typically uses reduced models based on acetylene species to predict the soot formation of different fuels

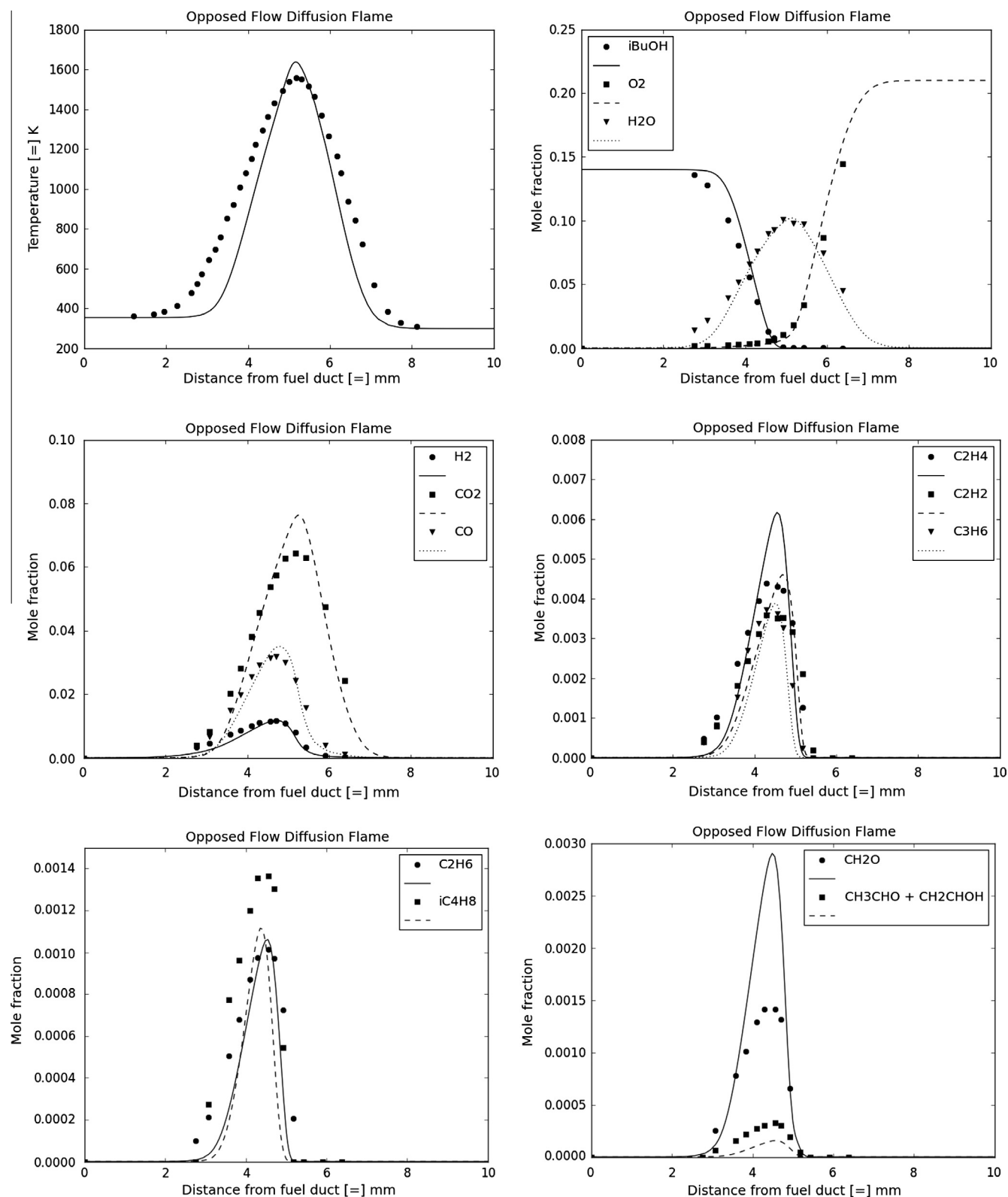


Fig. 17. Model comparison for all measured species in opposed flow diffusion flame of Grana et al. [16] (experimental – symbols and simulated – line).

in engines under different operating conditions. This approach will clearly lead to inaccurate results for modeling alternative fuels like *iso*-butanol.

The model also has reasonable predictions for most high temperature oxidation experiments. The current mechanism predicts the centerline concentration of all species formed in an *iso*-butanol

doped methane diffusion flame including aromatics such as benzene well. The model is able to accurately reproduce the peak flame speed at the correct stoichiometry (~ 1.1) and the flame speeds for the fuel lean case, but slightly underpredicts the flame speeds for the rich case. The model also predicts the concentration profiles of all major species in the opposed-flow diffusion flame

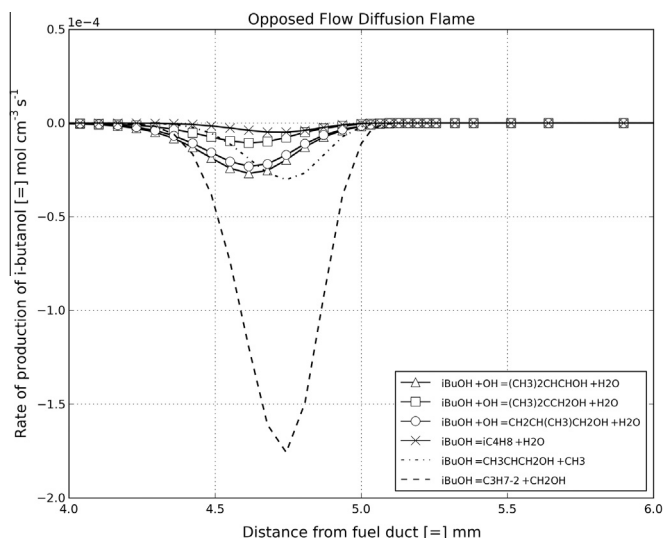


Fig. 18. Simulated rate of production for *iso*-butanol for opposed flow diffusion flame experiment [16].

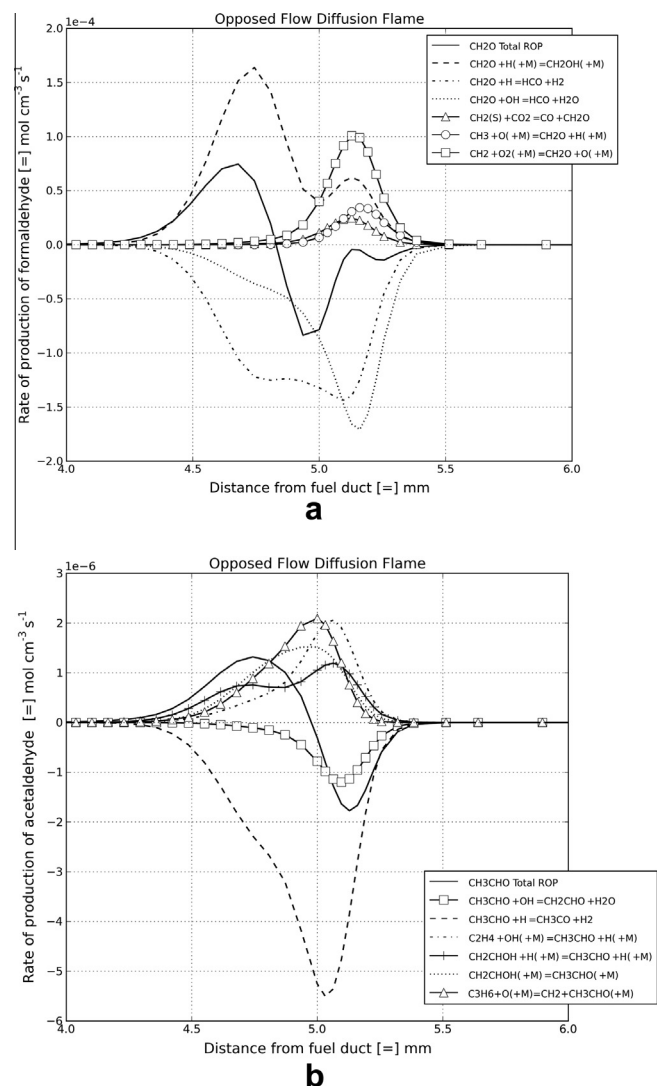


Fig. 19. Simulated rate of production for (a) formaldehyde and (b) acetaldehyde for opposed flow diffusion flame.

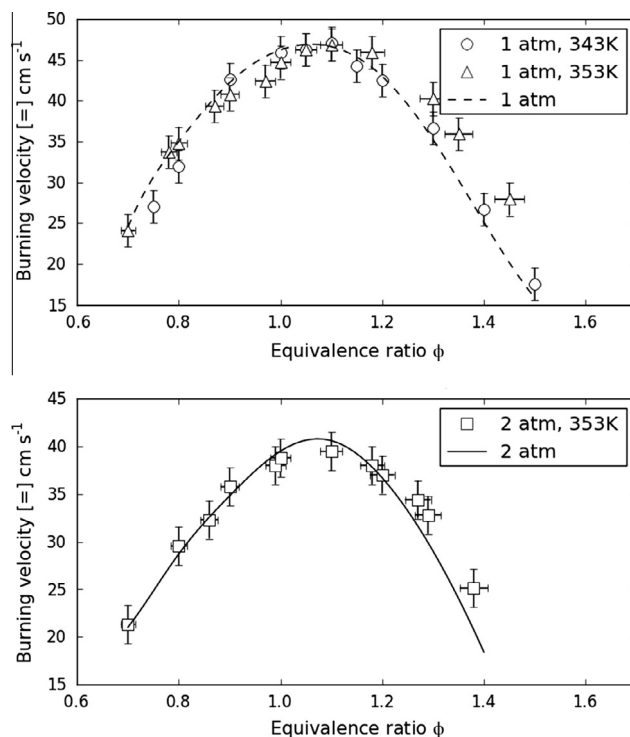


Fig. 20. Simulated (line) and experimental (symbol) zero-strain laminar flame speed of *iso*-butanol air mixtures as a function of equivalence ratios (Experiment: 1 atm $\circ$ Veloo et al. [12], Δ Liu et al. [13] and 2 atm $\square$ Liu et al. [13]).

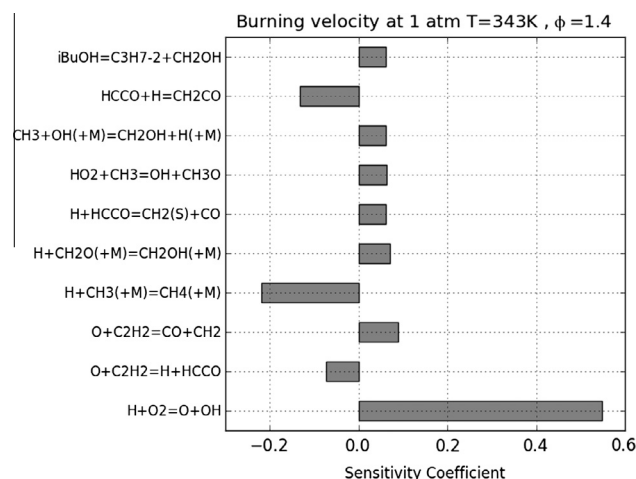


Fig. 21. Reaction sensitivity analysis for zero-strain laminar flame speed of *iso*-butanol in air mixtures, at $P = 1$ atm, $T(\text{unburnt}) = 343$ K and $\phi = 1.4$.

experiments measured by Grana et al. [16] well. Moreover, the calculated ignition delay times agree well with the ignition delays measured by Stranic et al. [17].

The performance for the jet stirred reactor experiments can be improved because the model significantly overpredicts the rate of consumption of *iso*-butanol especially for temperatures lower than 850 K. The discrepancy between the model and experiment can be attributed to the large uncertainties in calculated HO_2 + *iso*-butanol rate constants (since no hindered rotor were included in the rate calculation the uncertainty is an order of magnitude at $T \sim 800$ K) and in the estimated OH + *iso*-butanol product branching ratios. It is clear that having access to improved rate coefficients for the hydrogen abstraction reactions by OH and

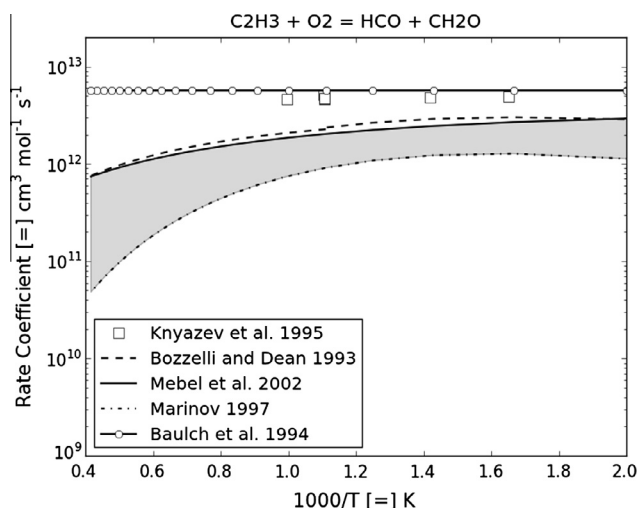


Fig. 22. Comparison of literature rate coefficients for $C_2H_3 + O_2 = HCO + CH_2O$ (Knyazev et al. [96] Bozzelli and Dean [98], Mebel et al. [99], Marinov [56] and Baulch et al. [103]). Present model uses Mebel et al. [99] value.

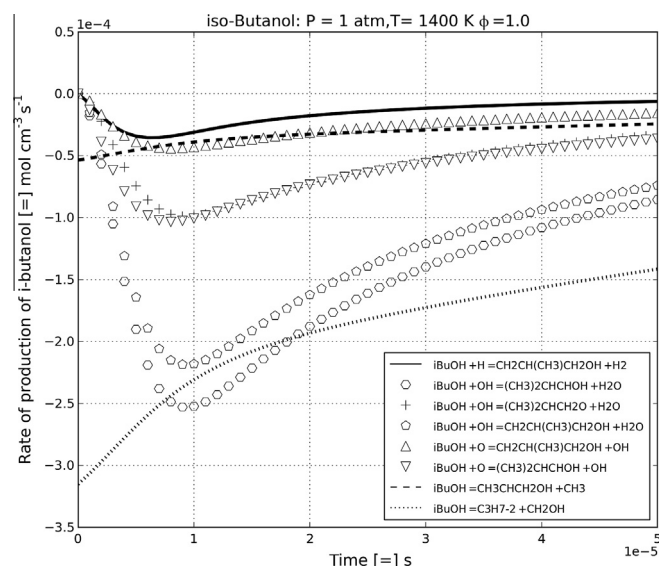


Fig. 24. Simulated rate of production of iso-butanol for adiabatic isochoric shock tube autoignition [17] at $P = 1$ atm, $T = 1400$ K and $\Phi = 1$.

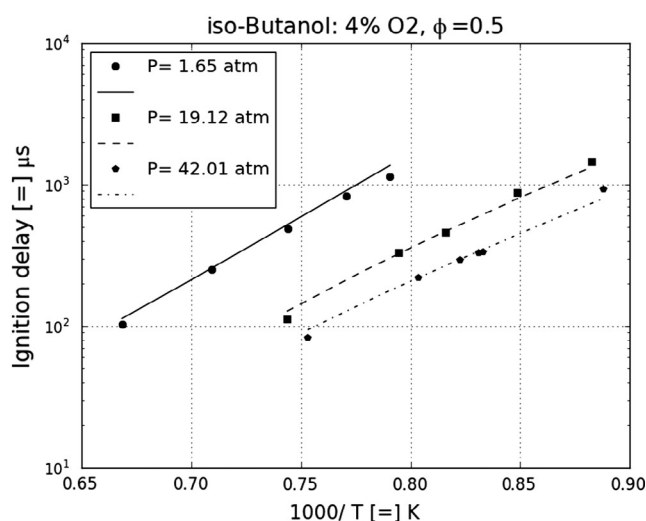
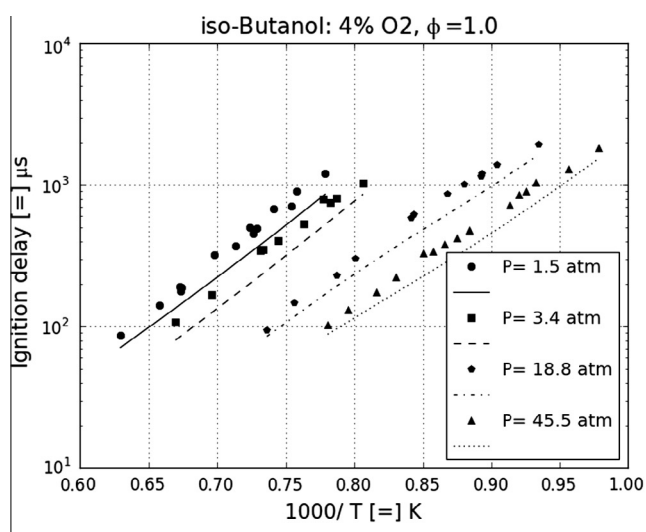


Fig. 23. Experimental [17] (shapes) and simulated (lines) autoignition delay for iso-butanol at high temperatures. The stated pressures are used for modeling the shock tube experiments.

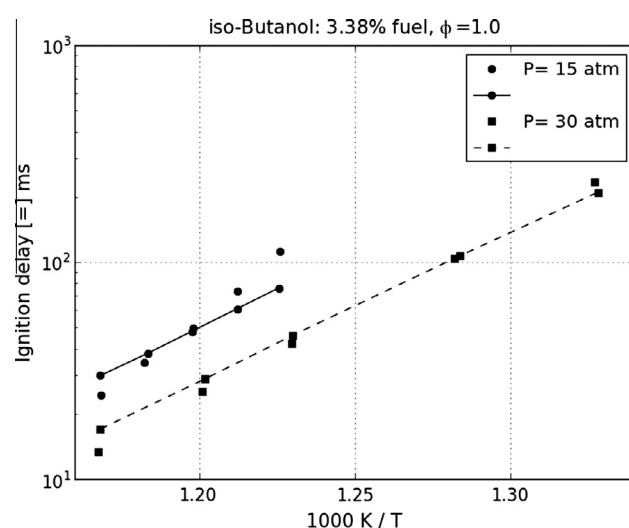


Fig. 25. Experimental [18] (symbols) and simulated (lines) autoignition delay for iso-butanol at low temperature. The stated pressure was used for modeling the RCM experiments. These simulations do not include the experimental heat losses, so the model vs data agreement is not as good at low temperature as it appears in this plot.

HO_2 with uncertainty lower than a factor 2 will be a significant step forward and allow development of better predictive kinetic models for alcohol oxidation. Another source of discrepancy between the model and experiments at low temperature is the peroxy chemistry as the current model rates are based on alkane-like rate constants and might have an incomplete treatment of all the pathways. We recommend the current model be used for applications where the temperature > 850 K. Further theoretical and quantum chemical studies are needed in understanding the rates of alcohol peroxy pathways and extending the model to lower temperatures.

Acknowledgments

The authors would like to thank Prof. Lisa Pfefferle and Dr. Charles McEnally for sharing their doped methane flame

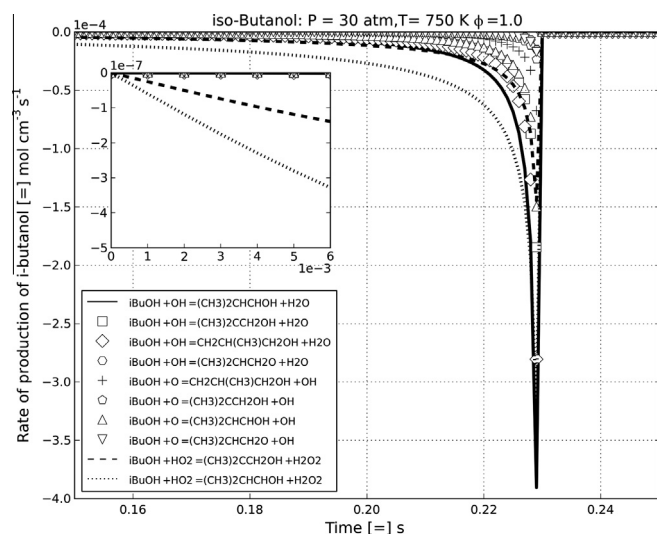


Fig. 26. Simulated rate of production of *iso*-butanol during adiabatic isochoric autoignition ($P = 30$ atm, $T = 750$ K and $\phi = 1$). Hydrogen abstraction by HO_2 is the dominant consumption path during the pre-ignition period.

experimental data. S.S.M. would like to thank Dr. Stephen Klippenstein for sharing the *iso*-butanol unimolecular decomposition rates and Dr. S. Mani Sarathy for helpful discussions. K.V.G. holds a Post-doctoral Fellowship of the Fund for Scientific Research, Flanders, Belgium. S.S.M., M.R.H. and W.H.G. would like to acknowledge the Energy Frontier Research Center for Combustion Science (Grant No. DE-SC0001198) supported by the US Department of Energy, Office of Basic Energy Sciences. R.L.S.'s work on the numerical solver and splitting methods for combustion simulations was supported by the US Department of Energy (Office of Basic Energy Sciences, Division of Chemical Sciences, Geosciences, and Biosciences, under Contract DE-FG02-98ER14914). E.F.Z. gratefully acknowledges CAPES Foundation (Ministry of Education of Brazil, Brasília, DF, Brazil) for support during his visit to MIT.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2013.04.023>.

References

- [1] P. Durre, J. Wiegel, R.J. Maier, M.W.W. Adams (Eds.), *Incredible Anaerobes: From Physiology to Genomics to Fuels*, vol. 1125, 2008, pp. 353–362.
- [2] A. Ranjan, V.S. Moholkar, *Int. J. Energy Res.* 36 (3) (2012) 277–323, <http://dx.doi.org/10.1002/er.1948>.
- [3] J. Swana, Y. Yang, M. Behnam, R. Thompson, *Bioresour. Technol.* 102 (2) (2011) 2112–2117, <http://dx.doi.org/10.1016/j.biortech.2010.08.051>.
- [4] P.S. Nigam, A. Singh, *Prog. Energy Combust. Sci.* 37 (1) (2011) 52–68, <http://dx.doi.org/10.1016/j.pecs.2010.01.003>.
- [5] T.C. Ezeji, N. Qureshi, H.P. Blaschek, *Curr. Opin. Biotechnol.* 18 (3) (2007) 220–227, <http://dx.doi.org/10.1016/j.copbio.2007.04.002>.
- [6] GEVO Inc. <<http://www.gevo.com/>>.
- [7] R. Kolodziej, J. Scheib, *Bio-isobutanol: The Next-generation Biofuel*. <<http://www.hydrocarbonprocessing.com/Article/3081332/Bio-isobutanol-The-next-generation-biofuel.html>>.
- [8] F.N. Alasfour, *Energy Sources* 21 (5) (1999) 379–394, <http://dx.doi.org/10.1080/00908319950014704>.
- [9] F.N. Alasfour, *Appl. Therm. Eng.* 18 (8) (1998) 609–618, [http://dx.doi.org/10.1016/s1359-4311\(97\)00082-3](http://dx.doi.org/10.1016/s1359-4311(97)00082-3).
- [10] M. Karabektas, M. Hosoz, *Renew. Energy* 34 (6) (2009) 1554–1559, <http://dx.doi.org/10.1016/j.renene.2008.11.003>.
- [11] M.I. Al-Hasan, M. Al-Momany, *Transport* 23 (4) (2008) 306–310, <http://dx.doi.org/10.3846/1648-4142.2008.23.306-310>.
- [12] P.S. Veloo, F.N. Egofoopoulos, *Proc. Combust. Inst.* 33 (1) (2011) 987–993, <http://dx.doi.org/10.1016/j.proci.2010.06.163>.
- [13] W. Liu, A.P. Kelley, C.K. Law, *Proc. Combust. Inst.* 33 (1) (2011) 995–1002, <http://dx.doi.org/10.1016/j.proci.2010.05.084>.
- [14] P. Osswald, H. Gueldenberg, K. Kohse-Hoinghaus, B. Yang, T. Yuan, F. Qi, *Combust. Flame* 158 (1) (2011) 2–15, <http://dx.doi.org/10.1016/j.combustflame.2010.06.003>.
- [15] C.S. McEnally, L.D. Pfefferle, *Proc. Combust. Inst.* 30 (1) (2005) 1363–1370, <http://dx.doi.org/10.1016/j.proci.2004.07.033>.
- [16] R. Grana, A. Frassoldati, T. Faravelli, U. Niemann, E. Ranzi, R. Seiser, R. Cattolica, K. Seshadri, *Combust. Flame* 157 (11) (2010) 2137–2154, <http://dx.doi.org/10.1016/j.combustflame.2010.05.009>.
- [17] I. Stranic, D.P. Chase, J.T. Harmon, S. Yang, D.F. Davidson, R.K. Hanson, *Combust. Flame* 159 (2) (2012) 516–527, <http://dx.doi.org/10.1016/j.combustflame.2011.08.014>.
- [18] B. Weber, C.-J. Sung, Paper 1B13, 7th US Nat. Combust. Inst., 2011.
- [19] B.W. Weber, C.-J. Sung, *Energy Fuels* (2013), <http://dx.doi.org/10.1021/ef302195c>.
- [20] C. Togbé, A. Mzé-Ahmed, P. Dagaut, *Energy Fuels* 24 (2010) 5244–5256, [10.1021/ef1008488](http://dx.doi.org/10.1021/ef1008488).
- [21] A. Frassoldati, R. Grana, T. Faravelli, E. Ranzi, P. Osswald, K. Kohse-Hoinghaus, *Combust. Flame* 159 (7) (2012) 2295–2311, <http://dx.doi.org/10.1016/j.combustflame.2012.03.002>.
- [22] S.M. Sarathy, S. Vranckx, K. Yasunaga, M. Mehl, P. Osswald, W.K. Metcalfe, C.K. Westbrook, W.J. Pitz, K. Kohse-Hoinghaus, R.X. Fernandes, H.J. Curran, *Combust. Flame* 159 (6) (2012) 2028–2055, <http://dx.doi.org/10.1016/j.combustflame.2011.12.017>.
- [23] J.T. Moss, A.M. Berkowitz, M.A. Oehlschlaeger, J. Biet, V. Warth, P.-A. Glaude, F. Battin-Leclerc, *J. Phys. Chem. A* 112 (43) (2008) 10843–10855, <http://dx.doi.org/10.1021/jp806464p>.
- [24] W. Tsang, 490a, *AIChE Annual Meeting*, 2010.
- [25] W.H. Green, J.W. Allen, R.W. Ashcraft, G.J. Beran, C.A. Class, C. Gao, C.F. Goldsmith, M.R. Harper, A. Jalan, G.R. Magoon, D.M. Matheu, S.S. Merchant, J.D. Mo, S. Petway, S. Raman, S. Sharma, J. Song, K.M. Van Geem, J. Wen, R.H. West, A. Wong, H.-W. Wong, P.E. Yelvington, J. Yu, RMG – Reaction Mechanism Generator v3.3. <<http://www.rmg.sourceforge.net/>>.
- [26] K.M. Van Geem, S.P. Pyl, G.B. Marin, M.R. Harper, W.H. Green, *Ind. Eng. Chem. Res.* 49 (21) (2010) 10399–10420, <http://dx.doi.org/10.1021/ie1005349>.
- [27] S.P. Pyl, C.M. Schietekat, K.M. Van Geem, M.-F. Reyniers, J. Vercammen, J. Beens, G.B. Marin, *J. Chromatogr. A* 1218 (21) (2011) 3217–3223.
- [28] M. Djokic, H.H. Carstensen, K.M. Van Geem, G.B. Marin, *Proc. Combust. Inst.* 34 (2013) 251–258, <http://dx.doi.org/10.1016/j.proci.2012.05.066>.
- [29] S.P. Pyl, K.M. Van Geem, P. Puimege, M.K. Sabbe, M.F. Reyniers, G.B. Marin, *Energy* 43 (1) (2012) 146–160, <http://dx.doi.org/10.1016/j.energy.2011.11.032>.
- [30] K.M. Van Geem, A. Cuoci, A. Frassoldati, S.P. Pyl, G.B. Marin, E. Ranzi, *Combust. Sci. Technol.* 184 (7–8) (2012) 942–955, [10.1080/00102202.2012.663987](http://dx.doi.org/10.1080/00102202.2012.663987).
- [31] CHEMKIN-Pro ver 5, Reaction Design Inc., San Diego, CA, 2011.
- [32] M.R. Harper, K.M. Van Geem, S.P. Pyl, G.B. Marin, W.H. Green, *Combust. Flame* 158 (1) (2011) 16–41, <http://dx.doi.org/10.1016/j.combustflame.2010.06.002>.
- [33] Q. Chen, G.F. Froment, *J. Anal. Appl. Pyrolysis* 21 (1–2) (1991) 27–50.
- [34] W.R. Herman, L. Leon, A.N. Russell, *Chemical Reactor Theory*, Prentice-Hall, Englewood Cliffs, NJ, 1977.
- [35] J.C. Lee, R.A. Yetter, F.L. Dryer, A.G. Tomboulides, S.A. Orszag, *Combust. Sci. Technol.* 159 (2000) 199–212.
- [36] A.H. Cutler, M.J. Antal, M. Jones, *Ind. Eng. Chem. Res.* 27 (4) (1988) 691–697.
- [37] S. Sharma, M.R. Harper, W.H. Green, *Combust. Flame* 157 (7) (2010) 1331–1345, <http://dx.doi.org/10.1016/j.combustflame.2010.02.012>.
- [38] B.A.V. Bennett, C.S. McEnally, L.D. Pfefferle, M.D. Smooke, *Combust. Flame* 123 (4) (2000) 522–546, [http://dx.doi.org/10.1016/s0010-2180\(00\)00158-9](http://dx.doi.org/10.1016/s0010-2180(00)00158-9).
- [39] C.S. McEnally, L.D. Pfefferle, *Combust. Flame* 152 (4) (2008) 469–481, <http://dx.doi.org/10.1016/j.combustflame.2007.11.003>.
- [40] G. Strang, *SIAM J. Numer. Anal.* 5 (3) (1968) 506–517.
- [41] R.L. Speth, W.H. Green, S. MacNamara, G. Strang, *SIAM J. Numer. Anal.*, submitted for publication.
- [42] D.R. Mott, E.S. Oran, B. van Leer, *J. Comput. Phys.* 164 (2) (2000) 407–428, <http://dx.doi.org/10.1006/jcph.2000.6605>.
- [43] A.C. Hindmarsh, P.N. Brown, K.E. Grant, S.L. Lee, R. Serban, D.E. Shumaker, C.S. Woodward, *ACM Trans. Math. Softw.* 31 (3) (2005) 363–396.
- [44] D.G. Goodwin, *Cantera: An Object-oriented Software Toolkit for Chemical Kinetics, Thermodynamics, and Transport Processes*. <<http://code.google.com/p/cantera/>>.
- [45] CHEMKIN-MFC ver 5, Reaction Design Inc., San Diego, CA, 2010.
- [46] J. Song, *Building Robust Chemical Reaction Mechanisms: Next Generation of Automatic Model Construction Software*, Massachusetts Institute of Technology, Cambridge, MA, 2004.
- [47] R.G. Susnow, A.M. Dean, W.H. Green, P. Peczak, L.J. Broadbelt, *J. Phys. Chem. A* 101 (20) (1997) 3731–3740, <http://dx.doi.org/10.1021/jp9637690>.
- [48] A.Y. Chang, J.W. Bozzelli, A.M. Dean, Z. Phys. Chem. 214 (11) (2000) 1533, <http://dx.doi.org/10.1524/zpch.2000.214.11.1533>.
- [49] N.J.B. Green, Z.A. Bhatti, *Phys. Chem. Chem. Phys.* 9 (31) (2007) 4275–4290.
- [50] J.A. Miller, S.J. Klippenstein, *J. Phys. Chem. A* 110 (36) (2006) 10528–10544, <http://dx.doi.org/10.1021/jp062693x>.
- [51] J.W. Allen, C.F. Goldsmith, W.H. Green, *Phys. Chem. Chem. Phys.* 14 (3) (2012) 1131–1155, <http://dx.doi.org/10.1039/c1cp22765c>.
- [52] S.W. Benson, *Thermochemical Kinetics: Methods for the Estimation of Thermochemical Data and Rate Parameters*, 1976.
- [53] J. Welty, C. Wicks, R. Wilson, G. Rorrer, *Fundamentals of Momentum, Heat, and Mass Transfer*, Wiley, 2000.
- [54] N. Hansen, M.R. Harper, W.H. Green, *Phys. Chem. Chem. Phys.* 13 (45) (2011) 20262–20274.

- [55] G.P. Smith, D.M. Golden, M. Frenklach, N.W. Moriarty, B. Eiteneer, M. Goldenberg, C.T. Bowman, R.K. Hanson, S. Song, W.C. Gardiner, V.V. Lissianski, Z. Qin, GRI-Mech 3.0. <http://www.me.berkeley.edu/gri_mech/>.
- [56] N.M. Marinov, *Int. J. Chem. Kinet.* 31 (3) (1999) 183–220.
- [57] M.P. Burke, M. Chaos, Y. Ju, F.L. Dryer, S.J. Klippenstein, *Int. J. Chem. Kinet.* (2012), <http://dx.doi.org/10.1002/kin.20603>.
- [58] C.-W. Zhou, S.J. Klippenstein, J.M. Simmie, H.J. Curran, *Proc. Combust. Inst.* (2012), <http://dx.doi.org/10.1016/j.proci.2012.06.034> (Available online 29 June 2012).
- [59] C. Eckart, *Phys. Rev.* 35 (11) (1930) 1303–1309.
- [60] C.M. Rosado-Reyes, W. Tsang, I.M. Alecu, S.S. Merchant, W.H. Green, unpublished result, 2012.
- [61] M. Yujing, A. Mellouki, *Chem. Phys. Lett.* 333 (1–2) (2001) 63–68, [http://dx.doi.org/10.1016/S0009-2614\(00\)01346-4](http://dx.doi.org/10.1016/S0009-2614(00)01346-4).
- [62] S.S. Vasu, D.F. Davidson, R.K. Hanson, D.M. Golden, *Chem. Phys. Lett.* 497 (1–3) (2010) 26–29, <http://dx.doi.org/10.1016/j.cplett.2010.08.001>.
- [63] G.A. Pang, R.K. Hanson, D.M. Golden, C.T. Bowman, *J. Phys. Chem. A* 116 (10) (2012) 2475–2483, <http://dx.doi.org/10.1021/jp211885p>.
- [64] C.-W. Zhou, J.M. Simmie, H.J. Curran, *Combust. Flame* 158 (4) (2011) 726–731, <http://dx.doi.org/10.1016/j.combustflame.2010.11.002>.
- [65] J. Zádor, J.A. Miller, 7th US Nat. Tech. Meet. Combust. Inst., 2011.
- [66] A. Mellouki, F. Oussar, X. Lun, A. Chakir, *Phys. Chem. Chem. Phys.* 6 (11) (2004) 2951–2955.
- [67] H.-H. Carstensen, A.M. Dean, O. Deutschmann, *Proc. Combust. Inst.* 31 (1) (2007) 149–157, <http://dx.doi.org/10.1016/j.proci.2006.08.091>.
- [68] E.S.C. Kwok, R. Atkinson, *Atmos. Environ.* 29 (14) (1995) 1685–1695, [http://dx.doi.org/10.1016/1352-2310\(95\)00069-B](http://dx.doi.org/10.1016/1352-2310(95)00069-B).
- [69] N. Cohen, *Int. J. Chem. Kinet.* 14 (12) (1982) 1339–1362, <http://dx.doi.org/10.1002/kin.550141206>.
- [70] J.J.A. Montgomery, M.J. Frisch, J.W. Ochterski, G.A. Petersson, *J. Chem. Phys.* 112 (15) (2000) 6532–6542.
- [71] J.J.A. Montgomery, J.W. Ochterski, G.A. Petersson, *J. Chem. Phys.* 101 (7) (1994) 5900–5909.
- [72] J.W. Ochterski, G.A. Petersson, J.J.A. Montgomery, *J. Chem. Phys.* 104 (7) (1996) 2598–2619.
- [73] M.J. Frisch, G.W. Trucks, H.B. Schlegel, G.E. Scuseria, M.A. Rob, J.R. Cheeseman, J.A. Montgomery Jr., K.N.K.T. Vreven, J.C. Burant, J.M. Millam, S.S. Iyengar, V.B.J. Tomasi, B. Mennucci, M. Cossi, G. Scalmani, N. Rega, G.A. Petersson, M.H.H. Nakatsuji, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, M. Klene, X. Li, J.E. Knox, H.P. Hratchian, J.B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R.E. Stratmann, O. Yazyev, A.J. Austin, R. Cammi, C. Pomelli, J.W. Ochterski, P.Y. Ayala, K. Morokuma, G.A. Voth, P. Salvador, J.J. Dannenberg, V.G. Zakrzewski, S. Dapprich, A.D. Daniels, M.C. Strain, O. Farkas, D.K. Malick, A.D. Rabuck, K. Raghavachari, J.B. Foresman, J.V. Ortiz, Q. Cui, A.G. Baboul, S. Clifford, J. Cioslowski, B.B. Stefanov, G. Liu, A. Liashenko, P. Piskorz, I. Komaromi, R.L. Martin, D.J. Fox, T. Keith, M.A. Al-Laham, C.Y. Peng, A. Nanayakkara, M. Challacombe, P.M.W. Gill, B. Johnson, W. Chen, M.W. Wong, C. Gonzalez, J.A. Pople, Gaussian 03, Wallingford, CT, 2003.
- [74] A.P. Scott, L. Radom, *J. Phys. Chem.* 100 (41) (1996) 16502–16513, <http://dx.doi.org/10.1021/jp960976r>.
- [75] G.A. Petersson, D.K. Malick, W.G. Wilson, J.W. Ochterski, J.J.A. Montgomery, M.J. Frisch, *J. Chem. Phys.* 109 (24) (1998) 10570–10579.
- [76] H.S. Johnston, J. Hecklen, *J. Phys. Chem.* 66 (3) (1962) 532–533, <http://dx.doi.org/10.1021/j100809a040>.
- [77] B.C. Garrett, D.G. Truhlar, *J. Phys. Chem.* 83 (22) (1979) 2921–2926, <http://dx.doi.org/10.1021/j100485a023>.
- [78] S. Sharma, M.R. Harper, W.H. Green, CANTHERM, Massachusetts Institute of Technology, 2012. <<https://github.com/GreenGroup/CanTherm>>.
- [79] M.L. Coote, *J. Phys. Chem. A* 108 (17) (2004) 3865–3872, <http://dx.doi.org/10.1021/jp049863v>.
- [80] C.F. Goldsmith, W.H. Green, S.J. Klippenstein, *J. Phys. Chem. A* 116 (13) (2012) 3325–3346, <http://dx.doi.org/10.1021/jp210722w>.
- [81] P. Seal, E. Papajak, D.G. Truhlar, *J. Phys. Chem. Lett.* 3 (2) (2012) 264–271, <http://dx.doi.org/10.1021/jz201546e>.
- [82] C.A. Taatjes, N. Hansen, A. McIlroy, J.A. Miller, J.P. Senosiain, S.J. Klippenstein, F. Qi, L.S. Sheng, Y.W. Zhang, T.A. Cool, J. Wang, P.R. Westmoreland, M.E. Law, T. Kasper, K. Kohse-Hoinghaus, *Science* 308 (5730) (2005) 1887–1889.
- [83] K.M. VanGeem, M.F. Reyniers, G.B. Marin, *Ind. Eng. Chem. Res.* 44 (10) (2005).
- [84] H. Richter, O.A. Mazyar, R. Sumathi, W.H. Green, J.B. Howard, J.W. Bozzelli, *J. Phys. Chem. A* 105 (9) (2001) 1561–1573.
- [85] C.S. McEnally, L.D. Pfefferle, *Combust. Flame* 148 (4) (2007) 210–222.
- [86] R.G. Butler, I. Glassman, *Proc. Combust. Inst.* 32 (2009) 395–402, <http://dx.doi.org/10.1016/j.proci.2008.05.010>.
- [87] M.K. Sabbe, K.M. Van Geem, M.F. Reyniers, G.B. Marin, *AIChE J.* 57 (2) (2011) 482–496, <http://dx.doi.org/10.1002/aic.12269>.
- [88] D. Wang, A. Violi, D.H. Kim, J.A. Mullholland, *J. Phys. Chem. A* 110 (14) (2006) 4719–4725, <http://dx.doi.org/10.1021/jp053628a>.
- [89] C. Cavallotti, D. Polino, A. Frassoldati, E. Ranzi, *J. Phys. Chem. A* 116 (13) (2012) 3313–3324, <http://dx.doi.org/10.1021/jp212151p>.
- [90] G. da Silva, C.-H. Kim, J.W. Bozzelli, *J. Phys. Chem. A* 110 (25) (2006) 7925–7934, <http://dx.doi.org/10.1021/jp0602878>.
- [91] J. Pardo, M.C. López, J. Santafé, F.M. Royo, J.S. Urieta, *Fluid Phase Equilib.* 119 (1–2) (1996) 165–173.
- [92] S. Dooley, M.P. Burke, M. Chaos, Y. Stein, F.L. Dryer, V.P. Zhukov, O. Finch, J.M. Simmie, H.J. Curran, *Int. J. Chem. Kinet.* 42 (9) (2010) 527–549, <http://dx.doi.org/10.1002/kin.20512>.
- [93] A.W. Jasper, S.J. Klippenstein, L.B. Harding, *Proc. Combust. Inst.* 32 (1) (2009) 279–286, <http://dx.doi.org/10.1016/j.proci.2008.05.036>.
- [94] B. Ruscic, Active Thermochemical Tables: version Alpha 1.110 of the Core (Argonne) Thermochemical Network, 04.02.2011.
- [95] J. Li, Z. Zhao, A. Kazakov, M. Chaos, F.L. Dryer, J.J. Scire, *Int. J. Chem. Kinet.* 39 (3) (2007) 109–136, <http://dx.doi.org/10.1002/kin.20218>.
- [96] V.D. Knyazev, I.R. Slagle, *J. Phys. Chem.* 99 (8) (1995) 2247–2249, <http://dx.doi.org/10.1021/j100008a002>.
- [97] A.J. Eskola, R.S. Timonen, *Phys. Chem. Chem. Phys.* 5 (12) (2003) 2557–2561.
- [98] J.W. Bozzelli, A.M. Dean, *J. Phys. Chem.* 97 (17) (1993) 4427–4441, <http://dx.doi.org/10.1021/j100119a030>.
- [99] A.M. Mebel, E.W.G. Diau, M.C. Lin, K. Morokuma, *J. Am. Chem. Soc.* 118 (40) (1996) 9759–9771, <http://dx.doi.org/10.1021/ja961476e>.
- [100] N.M. Marinov, W.J. Pitz, C.K. Westbrook, A.M. Vincitore, M.J. Castaldi, S.M. Senkan, C.F. Melius, *Combust. Flame* 114 (1–2) (1998) 192–213, [http://dx.doi.org/10.1016/S0010-2180\(97\)00275-7](http://dx.doi.org/10.1016/S0010-2180(97)00275-7).
- [101] S.M. Sarathy, M.J. Thomson, C. Togbé, P. Dagaut, F. Halter, C. Mounaim-Rousselle, *Combust. Flame* 156 (4) (2009) 852–864, <http://dx.doi.org/10.1016/j.combustflame.2008.11.019>.
- [102] H.J. Curran, W.J. Pitz, C.K. Westbrook, G.V. Callahan, F.L. Dryer, *Symp. Int. Combust. Proc.* 27 (1) (1998) 379–387, [http://dx.doi.org/10.1016/S0082-0784\(98\)80426-8](http://dx.doi.org/10.1016/S0082-0784(98)80426-8).
- [103] D.L. Baulch, C.J. Cobos, R.A. Cox, P. Frank, T.J.G. Hayman, T.M.J.A. Kerr, M.J. Pilling, J. Troe, R.W. Walker, J. Warnatz, *J. Phys. Chem. Ref. Data* 23 (1994) 847–1033.